

9. Clock Generation Circuit

9.1 Overview

This MCU incorporates a clock generation circuit.

Table 9.1 lists the specifications of the clock generation circuit. Figure 9.1 shows a block diagram of the clock generation circuit.

Table 9.1 Specifications of Clock Generation Circuit (1/2)

Item	Specification
Use	- Generates the system clock (ICLK) to be supplied to the CPU, DMAC, DTC, code flash memory, and RAM. - Generates the peripheral module clock (PCLKA) to be supplied to the ETHERC*1, EDMAC, EPTPC, RSPI, SCli, MTU, GLCDC, DRW2D, PMGI, and GPTW. - Generates the peripheral module clock (PCLKB) to be supplied to peripheral modules. - Generates the peripheral module clocks (for analog conversion) (PCLKC: unit 0; PCLKD: unit 1) to be supplied to S12AD. - Generates the flash-IF clock (FCLK) to be supplied to the flash interface. - Generates the external bus clock (BCLK) to be supplied to the external bus. - Generates the SDRAM clock (SDCLK) to be supplied to the SDRAM. - Generates the USB clock (UCLK) to be supplied to the USB. - Generates the CAC clock (CACCLK) to be supplied to the CAC. - Generates the CAN clock (CANMCLK) to be supplied to the CAN. - Generates the RTC sub-clock (RTCSCLK) to be supplied to the RTC. - Generates the RTC main clock (RTCMCLK) to be supplied to the RTC. - Generates the IWDG-dedicated clock (IWDTCCLK) to be supplied to the IWDG. - Generates the JTAG clock (JTAGTCK) to be supplied to the JTAG.
Operating frequency*2	- ICLK: 240 MHz (max)*3 - PCLKA: 120 MHz (max) - PCLKB: 60 MHz (max) - PCLKC: 60 MHz (max) - PCLKD: 60 MHz (max) - FCLK: 4 MHz to 60 MHz (for programming and erasing the code flash memory and data flash memory) 60 MHz (max) (for reading from the data flash memory) - BCLK: 120 MHz (max) - BCLK pin output: 80 MHz (max) - SDCLK pin output: 80 MHz (max) - UCLK: 48 MHz - CLKOUT25M pin output: 25 MHz - CLKOUT pin output: 40 MHz (max) - CACCLK: Same as the clock from respective oscillators. - CANMCLK: 24 MHz (max) - RTCSCLK: 32.768 kHz - RTCMCLK: 1 kHz to 16 MHz - IWDTCCLK: 120 kHz - JTAGTCK: 10 MHz (max)
Main clock oscillator	- Resonator frequency: 8 MHz to 24 MHz - External clock input frequency: 30 MHz (max) - Connectable resonator or additional circuit: ceramic resonator, crystal resonator - Connection pin: EXTAL, XTAL - Oscillation stop detection function: This function changes the system clock source to the LOCO and places the MTU and GPTW pins in the high-impedance state when stopping of oscillation by the main clock oscillator is detected.
Sub-clock oscillator	- Resonator frequency: 32.768 kHz - Connectable resonator or additional circuit: crystal resonator - Connection pin: XCIN, XCOUT
PLL frequency synthesizer (PLL)	- Input clock source: Main clock, HOCO - Input pulse frequency division ratio: Selectable from 1 to 3 - Input frequency: 8 MHz to 24 MHz - Frequency multiplication ratio: Selectable from 10 to 30 - Output clock frequency of the PLL frequency synthesizer: 120 MHz to 240 MHz

Table 9.1 Specifications of Clock Generation Circuit (2/2)

Item	Specification
PLL frequency synthesizer for specific purposes (PPLL)	• Input clock source: Main clock or HOCO • Input pulse frequency divisor: Selectable as from 1 to 3 • Input frequency: 8 MHz to 24 MHz • Multipliers: The multiplier is selectable as from 10 to 30 • Output clock frequency from the PLL frequency synthesizer: 120 MHz to 240 MHz
High-speed on-chip oscillator (HOCO)	• Selectable from 16 MHz, 18 MHz, and 20 MHz • HOCO power supply control
Low-speed on-chip oscillator (LOCO)	Oscillation frequency: 240 kHz
IWDT-dedicated on-chip oscillator	Oscillation frequency: 120 kHz
JTAG external clock input (TCK)	Input clock frequency: 10 MHz (max)
Control of output on the BCLK pin	• BCLK clock output or high output is selectable • BCLK or BCLK/2 is selectable
Control of output on the SDCLK pin	SDCLK clock output or high output is selectable
Event linking (output)	Detection of stopping of the main clock oscillator
Event linking (input)	Switching of the clock source to the low-speed on-chip oscillator

Note 1. Restrictions in relation to the clock when ETHERC is in use are as follows.

$$12.5 \text{ MHz} \leq \text{PCLKA} \leq 120 \text{ MHz}$$

Note 2. Restrictions on setting clock frequency: $\text{ICLK} \geq \text{BCLK}$, $\text{PCLKA} \geq \text{PCLKB}$, $\text{PCLKB} \geq \text{PCLKC}$, $\text{PCLKB} \geq \text{PCLKD}$

Restrictions on clock frequency ratio: (N: integer)

$$\text{ICLK:FCLK} = \text{N:1 or 1:N}; \text{ICLK:PCLKA} = \text{N:1 or 1:N}; \text{ICLK:PCLKB} = \text{N:1 or 1:N};$$

$$\text{ICLK:PCLKC} = \text{N:1 or 1:N}; \text{ICLK:PCLKD} = \text{N:1 or 1:N}; \text{ICLK:BCLK} = \text{N:1}$$

Note 3. When the frequency of ICLK is set to faster than 120 MHz, the value of the MEMWAIT register needs to be modified.

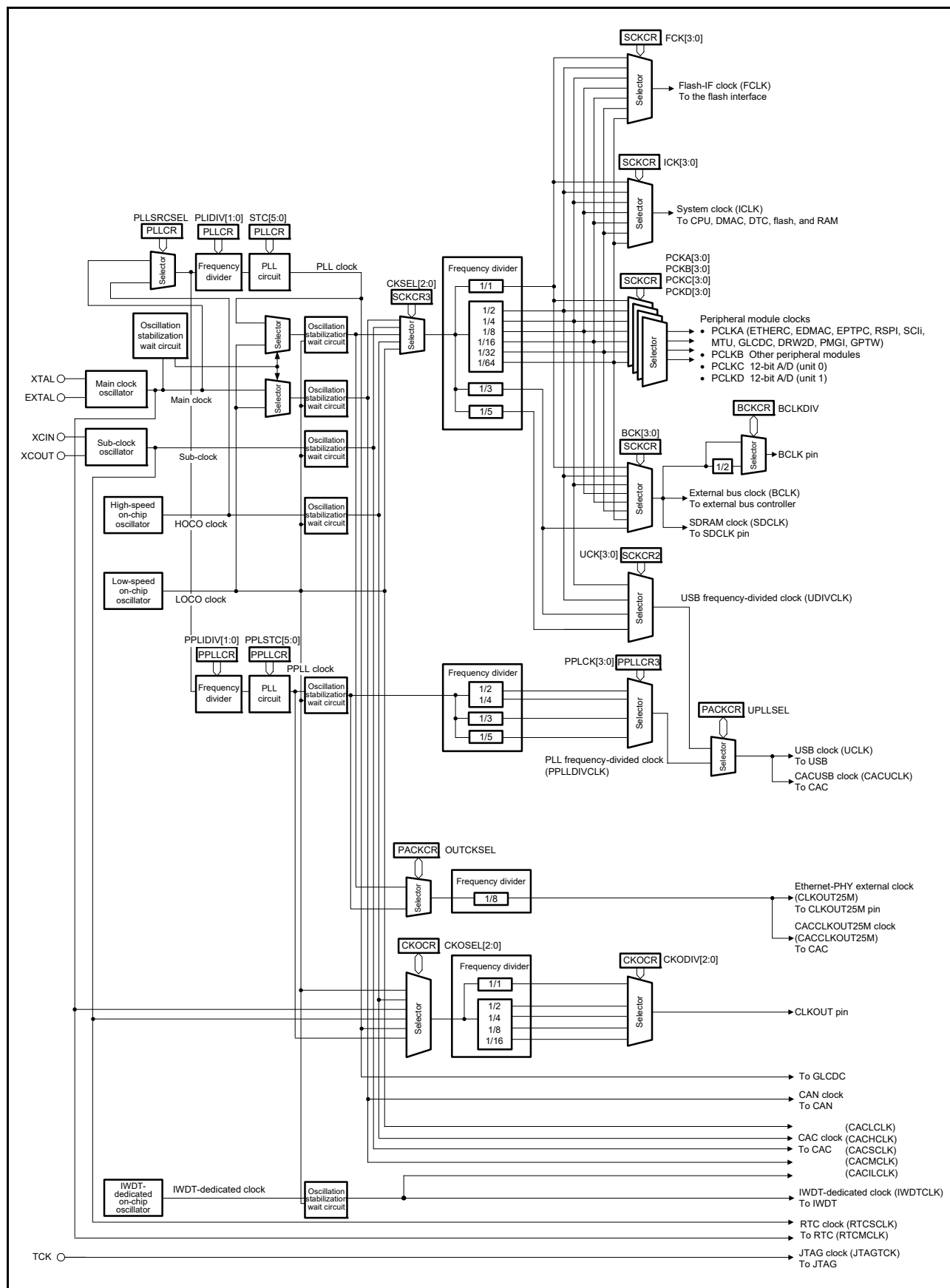


Figure 9.1 Block Diagram of Clock Generation Circuit

Table 9.2 lists the input/output pins of the clock generation circuit.

Table 9.2 Input/Output Pins of Clock Generation Circuit

Pin Name	I/O	Description
XTAL	Output	These pins are used to connect a crystal resonator. The EXTAL pin can also be used to input an external clock. For details, section 9.3.2, External Clock Input.
EXTAL	Input	
XCIN	Input	These pins are used to connect a 32.768-kHz crystal resonator.
XCOUT	Output	
TCK	Input	This pin is used to input the clock for the JTAG.
BCLK	Output	This pin is used to supply external devices with the external bus clock (BCLK).
SDCLK	Output	This pin is used to supply external devices with the SDRAM clock (SDCLK).
CLKOUT	Output	This is the CLKOUT output pin.
CLKOUT25M	Output	This pin provides the Ethernet-PHY external clock (CLKOUT25M) to the external chip.

9.2 Register Descriptions

9.2.1 System Clock Control Register (SCKCR)

Address(es): 0008 0020h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	FCK[3:0]				ICK[3:0]				PSTOP ₁	PSTOP ₀	—	—	BCK[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	PCKA[3:0]				PCKB[3:0]				PCKC[3:0]				PCKD[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	PCKD[3:0]	Peripheral Module Clock D (PCLKD) Select	b3 b0 0 0 0 0: ×1/1 0 0 0 1: ×1/2 0 0 1 0: ×1/4 0 0 1 1: ×1/8 0 1 0 0: ×1/16 0 1 0 1: ×1/32 0 1 1 0: ×1/64 Settings other than above are prohibited.	R/W
b7 to b4	PCKC[3:0]	Peripheral Module Clock C (PCLKC) Select	b7 b4 0 0 0 0: ×1/1 0 0 0 1: ×1/2 0 0 1 0: ×1/4 0 0 1 1: ×1/8 0 1 0 0: ×1/16 0 1 0 1: ×1/32 0 1 1 0: ×1/64 Settings other than above are prohibited.	R/W
b11 to b8	PCKB[3:0]	Peripheral Module Clock B (PCLKB) Select	b11 b8 0 0 0 0: ×1/1 0 0 0 1: ×1/2 0 0 1 0: ×1/4 0 0 1 1: ×1/8 0 1 0 0: ×1/16 0 1 0 1: ×1/32 0 1 1 0: ×1/64 Settings other than above are prohibited.	R/W
b15 to b12	PCKA[3:0]	Peripheral Module Clock A (PCLKA) Select	b15 b12 0 0 0 0: ×1/1 0 0 0 1: ×1/2 0 0 1 0: ×1/4 0 0 1 1: ×1/8 0 1 0 0: ×1/16 0 1 0 1: ×1/32 0 1 1 0: ×1/64 Settings other than above are prohibited.	R/W
b19 to b16	BCK[3:0]	External Bus Clock (BCLK) Select*1, *2	b19 b16 0 0 0 0: ×1/1 0 0 0 1: ×1/2 0 0 1 0: ×1/4 0 0 1 1: ×1/8 0 1 0 0: ×1/16 0 1 0 1: ×1/32 0 1 1 0: ×1/64 1 0 0 1: ×1/3 Settings other than above are prohibited.	R/W

Bit	Symbol	Bit Name	Description	R/W
b21, b20	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b22	PSTOP0	SDCLK Pin Output Control	0: SDCLK pin output is enabled. 1: SDCLK pin output is disabled. (Fixed high)	R/W
b23	PSTOP1	BCLK Pin Output Control*3	0: BCLK pin output is enabled. 1: BCLK pin output is disabled. (Fixed high)	R/W
b27 to b24	ICK[3:0]	System Clock (ICLK) Select *1, *2, *4	b27 b24 0 0 0 0: $\times 1/1$ 0 0 0 1: $\times 1/2$ 0 0 1 0: $\times 1/4$ 0 0 1 1: $\times 1/8$ 0 1 0 0: $\times 1/16$ 0 1 0 1: $\times 1/32$ 0 1 1 0: $\times 1/64$ Settings other than above are prohibited.	R/W
b31 to b28	FCK[3:0]	Flash-IF Clock (FCLK) Select *4	b31 b28 0 0 0 0: $\times 1/1$ 0 0 0 1: $\times 1/2$ 0 0 1 0: $\times 1/4$ 0 0 1 1: $\times 1/8$ 0 1 0 0: $\times 1/16$ 0 1 0 1: $\times 1/32$ 0 1 1 0: $\times 1/64$ Settings other than above are prohibited.	R/W

Note 1. Set the bits so that the frequency ratio of ICLK to BCLK becomes N to 1, where N is an integer.

Note 2. Do not make a setting such that the ICLK runs at a lower frequency than the external bus clock.

Note 3. When operation of the external bus clock is selected, the P53 I/O port pin function is not available because it is multiplexed on the same pin as the BCLK pin function.

Note 4. When the SCKCR3.CKSEL[2:0] bits are selecting the sub-clock oscillator in low-speed operating mode 2, division by one is the only frequency division setting allowed for the ICLK and FCLK.

SCKCR should not be modified in the following cases:

- The operating power control mode transition status flag in the operating power control register (OPCCR.OPCMTSF) is 1 (a transition to operating power control mode in progress)
- The code flash memory P/E mode entry bit or the data flash memory P/E mode entry bit in the flash P/E mode entry register (FENTRYR.FENTRYC or FENTRYR.FENTRYD) is 1 (P/E mode)
- Time period from WAIT instruction execution for a transition to sleep mode, to return from sleep mode to normal operating mode

9.2.2 Memory Wait Cycle Setting Register (MEMWAIT)

Address(es): 0008 101Ch

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	MEMWAIT
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	MEMWAIT	Memory Wait Cycle Setting	0: No wait 1: One wait cycle	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The MEMWAIT register specifies the number of wait cycles for access to the code flash memory, ECCRAM, and extended RAM.

For the number of wait cycles to the flash memory, refer to [section 62, Flash Memory \(FLASH\)](#); and for the number of wait cycles to the ECCRAM and extended RAM, refer to [section 60, RAM](#). There are no cycles of waiting time for access to the normal RAM regardless of the value of the MEMWAIT bit.

Set the MEMWAIT to 1 (one wait cycle) if the ICLK frequency is to be above 120 MHz.

Table 9.3 lists the restrictions for setting the MEMWAIT register.

Table 9.3 Restrictions on Setting of the MEMWAIT Register

MEMWAIT Bit	ICLK ≤ 120 MHz	ICLK > 120 MHz
0	Setting possible	Setting prohibited
1	Setting possible	Setting possible

To change the frequency of the ICLK from 120 MHz or less to above 120 MHz, start by setting the MEMWAIT bit to 1 (one wait cycle), and then change the frequency setting in the SCKCR register. To change the frequency from above 120 MHz to 120 MHz or less, start by changing the frequency setting to 120 MHz or less in the SCKCR register, and then set the MEMWAIT bit to 0 (no wait cycles).

Figure 9.2 and Figure 9.3 show the procedure for modifying the MEMWAIT bit.

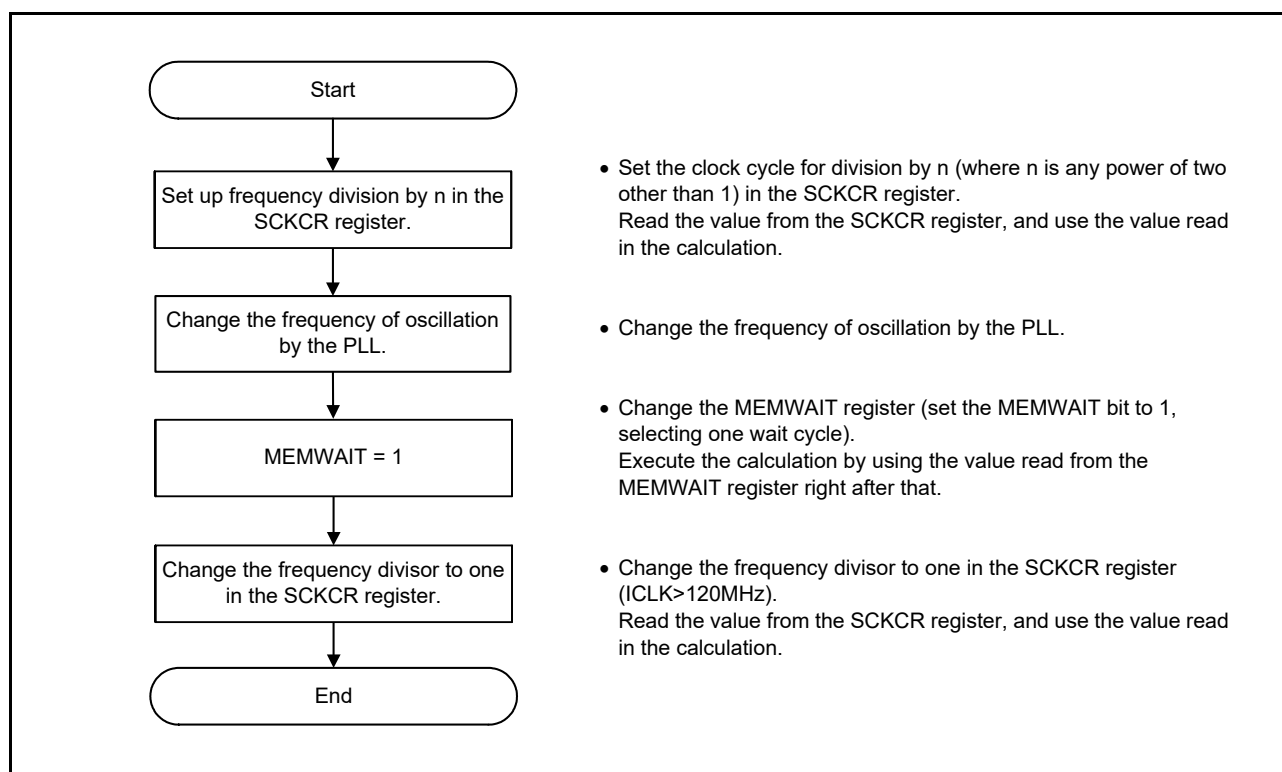


Figure 9.2 Example of Procedure for Changing Conditions from ICLK being at or below 120 MHz and MEMWAIT being 0 to ICLK being above 120 MHz and MEMWAIT being 1

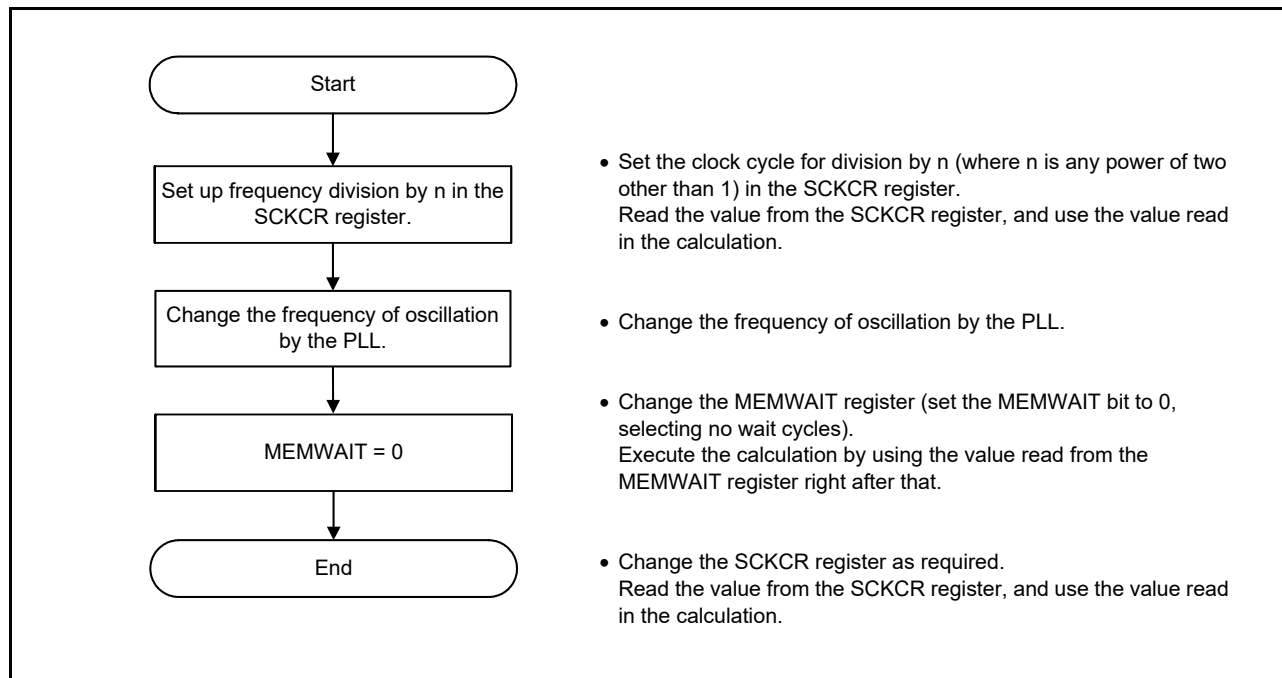


Figure 9.3 Example of Procedure for Changing Conditions from ICLK being above 120 MHz and MEMWAIT being 0 to ICLK being at or below 120 MHz and MEMWAIT being 1

9.2.3 System Clock Control Register 2 (SCKCR2)

Address(es): 0008 0024h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	—	UCLK[3:0]			—	—	—	—	—
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b3 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b7 to b4	UCLK[3:0]	USB Clock (UCLK) Select	b7b4 0001: ×1/2 0010: ×1/3 0011: ×1/4 0100: ×1/5 Settings other than above are prohibited.*1	R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Settings other than above are prohibited when USB is in use. When USB is not in use, these bits are read as 0001b. The write value should be 0001b.

SCKCR2 should not be modified in the following cases:

- The operating power control mode transition status flag in the operating power control register (OPCCR.OPCMTSF) is 1 (a transition to operating power control mode in progress)
- The code flash memory P/E mode entry bit or the data flash memory P/E mode entry bit in the flash P/E mode entry register (FENTRYR.FENTRYC or FENTRYR.FENTRYD) is 1 (P/E mode)
- Time period from WAIT instruction execution for a transition to sleep mode, to return from sleep mode to normal operating mode

UCLK[3:0] Bits (USB Clock (UCLK) Select)

These bits select the frequency of the USB clock (UCLK). The bits are effective when the USB clock source selection bit (PACKCR.UPLLSEL) is 0 (selecting the PLL). Set the frequency in the PPLLCR3 register while the USB clock source selection bit (PACKCR.UPLLSEL) is 1 (selecting the PPLL).

To specify the frequency in the PPLLCR3 register, refer to section 9.2.25, PPLL Control Register 3 (PPLLCR3).

The duty ratio is 2:1 when × 1/3 is selected while it is 3:2 when × 1/5 is selected.

9.2.4 System Clock Control Register 3 (SCKCR3)

Address(es): 0008 0026h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	CKSEL[2:0]		—	—	—	—	—	—	—	—	—
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b10 to b8	CKSEL[2:0]	Clock Source Select	b10 b8 0 0 0: LOCO 0 0 1: HOCO 0 1 0: Main clock oscillator 0 1 1: Sub-clock oscillator 1 0 0: PLL circuit Settings other than above are prohibited.	R/W
b15 to b11	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

SCKCR3 should not be modified in the following cases:

- The operating power control mode transition status flag in the operating power control register (OPCCR.OPCMTSF) is 1 (a transition to operating power control mode in progress)
- The code flash memory P/E mode entry bit or the data flash memory P/E mode entry bit in the flash P/E mode entry register (FENTRYR.FENTRYC or FENTRYR.FENTRYD) is 1 (P/E mode)
- Time period from WAIT instruction execution for a transition to sleep mode, to return from sleep mode to normal operating mode

CKSEL[2:0] Bits (Clock Source Select)

For the system clock (ICLK), peripheral module clocks (PCLKA, PCLKB, PCLKC, and PCLKD), flash interface clock (FCLK), external bus clock (BCLK), SDRAM clock (SDCLK), and USB clock source select bit (PACKCR.UPLLSEL) set to 0 (selecting the PLL), select the clock source from among the low speed on-chip oscillator (LOCO), high-speed on-chip oscillator (HOCO), main clock oscillator, sub-clock oscillator, or PLL circuit.

Transitions to clock sources which are not in operation are prohibited.

9.2.5 PLL Control Register (PLLCR)

Address(es): 0008 0028h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	STC[5:0]					—	—	—	PLLSR CSEL	—	—	PLIDIV[1:0]		
Value after reset:															
0	0	0	1	1	1	0	1	0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W																																																																																										
b1, b0	PLIDIV[1:0]	PLL Input Frequency Division Ratio Select	b1 b0 0 0: ×1/1 0 1: ×1/2 1 0: ×1/3 1 1: Setting prohibited	R/W																																																																																										
b3, b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W																																																																																										
b4	PLLSRCSEL	PLL Clock Source Select *1	0: Main clock oscillator 1: HOCO	R/W																																																																																										
b7 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W																																																																																										
b13 to b8	STC[5:0]	Frequency Multiplication Factor Select	<table><tr><td>b13</td><td>b8</td><td>b13</td><td>b8</td><td>b13</td><td>b8</td></tr><tr><td>0 1 0 0 1 1</td><td>×10.0</td><td>1 0 0 0 0 1</td><td>×17.0</td><td>1 0 1 1 1 1</td><td>×24.0</td></tr><tr><td>0 1 0 1 0 0</td><td>×10.5</td><td>1 0 0 0 1 0</td><td>×17.5</td><td>1 1 0 0 0 0</td><td>×24.5</td></tr><tr><td>0 1 0 1 0 1</td><td>×11.0</td><td>1 0 0 0 1 1</td><td>×18.0</td><td>1 1 0 0 0 1</td><td>×25.0</td></tr><tr><td>0 1 0 1 1 0</td><td>×11.5</td><td>1 0 0 1 0 0</td><td>×18.5</td><td>1 1 0 0 1 0</td><td>×25.5</td></tr><tr><td>0 1 0 1 1 1</td><td>×12.0</td><td>1 0 0 1 0 1</td><td>×19.0</td><td>1 1 0 0 1 1</td><td>×26.0</td></tr><tr><td>0 1 1 0 0 0</td><td>×12.5</td><td>1 0 0 1 1 0</td><td>×19.5</td><td>1 1 0 1 0 0</td><td>×26.5</td></tr><tr><td>0 1 1 0 0 1</td><td>×13.0</td><td>1 0 0 1 1 1</td><td>×20.0</td><td>1 1 0 1 0 1</td><td>×27.0</td></tr><tr><td>0 1 1 0 1 0</td><td>×13.5</td><td>1 0 1 0 0 0</td><td>×20.5</td><td>1 1 0 1 1 0</td><td>×27.5</td></tr><tr><td>0 1 1 0 1 1</td><td>×14.0</td><td>1 0 1 0 0 1</td><td>×21.0</td><td>1 1 0 1 1 1</td><td>×28.0</td></tr><tr><td>0 1 1 1 0 0</td><td>×14.5</td><td>1 0 1 0 1 0</td><td>×21.5</td><td>1 1 1 0 0 0</td><td>×28.5</td></tr><tr><td>0 1 1 1 0 1</td><td>×15.0</td><td>1 0 1 0 1 1</td><td>×22.0</td><td>1 1 1 0 0 1</td><td>×29.0</td></tr><tr><td>0 1 1 1 1 0</td><td>×15.5</td><td>1 0 1 1 0 0</td><td>×22.5</td><td>1 1 1 0 1 0</td><td>×29.5</td></tr><tr><td>0 1 1 1 1 1</td><td>×16.0</td><td>1 0 1 1 0 1</td><td>×23.0</td><td>1 1 1 0 1 1</td><td>×30.0</td></tr><tr><td>1 0 0 0 0 0</td><td>×16.5</td><td>1 0 1 1 1 0</td><td>×23.5</td><td></td><td></td></tr></table> Settings other than above are prohibited.	b13	b8	b13	b8	b13	b8	0 1 0 0 1 1	×10.0	1 0 0 0 0 1	×17.0	1 0 1 1 1 1	×24.0	0 1 0 1 0 0	×10.5	1 0 0 0 1 0	×17.5	1 1 0 0 0 0	×24.5	0 1 0 1 0 1	×11.0	1 0 0 0 1 1	×18.0	1 1 0 0 0 1	×25.0	0 1 0 1 1 0	×11.5	1 0 0 1 0 0	×18.5	1 1 0 0 1 0	×25.5	0 1 0 1 1 1	×12.0	1 0 0 1 0 1	×19.0	1 1 0 0 1 1	×26.0	0 1 1 0 0 0	×12.5	1 0 0 1 1 0	×19.5	1 1 0 1 0 0	×26.5	0 1 1 0 0 1	×13.0	1 0 0 1 1 1	×20.0	1 1 0 1 0 1	×27.0	0 1 1 0 1 0	×13.5	1 0 1 0 0 0	×20.5	1 1 0 1 1 0	×27.5	0 1 1 0 1 1	×14.0	1 0 1 0 0 1	×21.0	1 1 0 1 1 1	×28.0	0 1 1 1 0 0	×14.5	1 0 1 0 1 0	×21.5	1 1 1 0 0 0	×28.5	0 1 1 1 0 1	×15.0	1 0 1 0 1 1	×22.0	1 1 1 0 0 1	×29.0	0 1 1 1 1 0	×15.5	1 0 1 1 0 0	×22.5	1 1 1 0 1 0	×29.5	0 1 1 1 1 1	×16.0	1 0 1 1 0 1	×23.0	1 1 1 0 1 1	×30.0	1 0 0 0 0 0	×16.5	1 0 1 1 1 0	×23.5			R/W
b13	b8	b13	b8	b13	b8																																																																																									
0 1 0 0 1 1	×10.0	1 0 0 0 0 1	×17.0	1 0 1 1 1 1	×24.0																																																																																									
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b15, b14	—	Reserved	These bits are read as 0. The write value should be 0.	R/W																																																																																										

Note 1. Set this bit to 0 when using the USB.

Writing to the PLLCR is prohibited when the PLLCR2.PPLEN bit is 0 (the PLL operates).

In addition, writing to the PLLCR.PLLSRCSEL bit while the PLLCR2.PPLEN bit is 0 (PLL is operating) or the PPLLCR2.PPLEN bit is 0 (PPLL is operating) is prohibited.

PLIDIV[1:0] Bits (PLL Input Frequency Division Ratio Select)

These bits select the frequency division ratio of the PLL clock source.

Set these bits so that the frequency of PLL input signal is within the range of 8 MHz to 24 MHz.

PLLSRCSEL Bit (PLL Clock Source Select)

This bit selects the clock source for the PLL and PPLL.

STC[5:0] Bits (Frequency Multiplication Factor Select)

These bits select the frequency multiplication factor of the PLL circuit.

Set these bits so that the output frequency is within the range of the output clock frequency of the PLL circuit (120 MHz to 240 MHz).

9.2.6 PLL Control Register 2 (PLLCR2)

Address(es): 0008 002Ah

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	PLLEN
0	0	0	0	0	0	0	1

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	PLLEN	PLL Stop Control	0: PLL is operating. 1: PLL is stopped.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

PLLEN Bit (PLL Stop Control)

This bit runs or stops the PLL circuit.

The PLL clock source is selectable as the main clock oscillator and HOCO.

Selecting the main clock oscillator as the PLL clock source with the PLLCR.PLLSRCSEL bit requires setting the main clock oscillator wait control register (MOSCWTCR).

After the setting of the PLLEN bit has been changed to make the PLL run, only start using the PLL clock after confirming that the OSCOVFSR.PLOVF flag has been set to 1.

That is, a fixed time for stabilization is required after the setting for PLL operation. A fixed time is also required for oscillation to stop after the setting to stop PLL operation. Accordingly, take note of the following limitations when starting and stopping PLL operation by the PLLEN bit. The following notes apply when selecting the main clock oscillator as the PLL clock source.

- Setting PLL operation with the PLLEN bit is possible regardless of the setting of the OSCOVFSR.PLOVF flag. However, the time until deactivation of the PLL is completed (the time until the PLOVF flag is set to 0 after the setting to stop PLL operation) means that writing to the PLLCR2 register takes longer than the setting to operate the PLL.
- The PLL can be stopped by the PLLEN bit regardless of the setting of the OSCOVFSR.PLOVF flag. However, the wait for oscillation to become stable, i.e. the oscillation settling time (the time until the PLOVF flag is set to 1 after the setting to operate the PLL), means that writing to the PLLCR2 register takes longer than the setting to stop PLL operation.
- Regardless of whether or not the PLL clock is selected as the system clock, confirm that the OSCOVFSR.PLOVF flag has been set to 1 before executing a WAIT instruction to place the chip to software standby or deep software standby after the setting to operate the PLL.
- When a transition to software standby or deep software standby is to follow the setting to stop the PLL, confirm that the OSCOVFSR.PLOVF flag has been set to 0 before executing the WAIT instruction.

Writing of 1 to the PLLEN bit (stopping the PLL) is prohibited while the PLL clock is selected by the clock source select bits in system clock control register 3 (SCKCR3.CKSEL[2:0]).

Writing of 0 to the PLLEN bit (making the PLL operate) is prohibited when the setting of the operating power control mode select bits in the operating power control register (OPCCR.OPCM[2:0]) is for low-speed operating mode 1 or 2.

9.2.7 External Bus Clock Control Register (BCKCR)

Address(es): 0008 0030h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	BCLKDIV
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	BCLKDIV	BCLK Pin Output Select	0: BCLK 1: 1/2 BCLK	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

BCKCR should not be modified in the following cases:

- The operating power control mode transition status flag in the operating power control register (OPCCR.OPCMTSF) is 1 (a transition to operating power control mode in progress)
- The code flash memory P/E mode entry bit or the data flash memory P/E mode entry bit in the flash P/E mode entry register (FENTRYR.FENTRYC or FENTRYR.FENTRYD) is 1 (P/E mode).
- Time period from WAIT instruction execution for a transition to sleep mode, to return from sleep mode to normal operating mode

BCLKDIV Bit (BCLK Pin Output Select)

This bit selects the clock signal for output from the BCLK pin.

Either the BCLK clock with the frequency selected by the BCK[3:0] bits in SCKCR or the BCLK clock divided by 2 can be selected. To control external bus control signals at the falling edge of the BCLK pin, set this bit to 1.

9.2.8 Main Clock Oscillator Control Register (MOSCCR)

Address(es): 0008 0032h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	MOSTP
0	0	0	0	0	0	0	1

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	MOSTP	Main Clock Oscillator Stop	0: Main clock oscillator is operating. 1: Main clock oscillator is stopped.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

MOSTP Bit (Main Clock Oscillator Stop)

This bit runs or stops the main clock oscillator.

The main clock oscillator is operated or stopped by the MOSTP bit and main clock oscillator forced oscillation bit in the main clock oscillator forced oscillation control register (MOFCR.MOFXIN). The main clock oscillator can be started by setting the MOSTP bit to operating or by setting the MOFXIN bit to forced oscillation. When the MOFXIN bit is set to forced oscillation, the oscillator operates even in software standby mode and deep software standby mode.

When the main clock is to be used as the source to drive counting by the realtime clock, use the MOFCR.MOFXIN bit to make the main clock run. If the main clock is only to be used as the system clock or PLL clock source, use the MOSCCR.MOSTP bit.

When the main clock is to be used as the clock source for counting and also for another purpose, use both the MOFCR.MOFXIN and MOSCCR.MOSTP bits to make the main clock run.

When changing the value of the MOSTP bit or MOFCR.MOFXIN bit, execute subsequent instructions after reading the bit and checking that its value has actually been updated (refer to (2) Notes on Writing to I/O Registers, in section 5, I/O Registers).

When the main clock is not to be used as the clock source for counting by the realtime clock, the main clock oscillator wait control register (MOSCWTCR) must be set. In this case, after the setting of the MOSCCR.MOSTP bit has been changed to make the main clock run, only start using the main clock after confirming that the OSCOVFSR.MOOVF flag has been set to 1.

When the main clock is to be used as the clock source for counting by the realtime clock, after the setting of MOFCR.MOFXIN bit has been changed to make the main clock run, only start using the main clock as the clock source for counting by the realtime clock after software has checked that the main clock oscillation stabilization time (crystal) (tMAINOSC) has elapsed. When an external clock signal is to be input for supply to the main clock oscillator, waiting for the main clock oscillation stabilization time (tMAINOSC) is not necessary.

For the main clock oscillator, a fixed time is required for oscillation to become stable after the settings for operation have been made. Furthermore, a fixed time is required for oscillation to actually stop after the settings to stop oscillation have been made. Accordingly, take note of the following limitations when starting and stopping operation.

- Main-clock operation can be selected with the MOSTP bit regardless of the setting of the OSCOVFSR.MOOVF flag. However, the time until deactivation of the main clock is completed (the time until the OSCOVFSR.MOOVF flag is set to 0 after the setting to stop operation) means that writing to the MOSCCR register takes longer than the setting to operate the main clock.
- The main clock can be stopped by the MOSTP bit regardless of the setting of the OSCOVFSR.MOOVF flag. However, the wait for oscillation to become stable, i.e. the oscillation settling time (the time until the OSCOVFSR.MOOVF flag is set to 1 after the setting to operate the main clock), means that writing to the MOSCCR register takes longer than the setting to stop operation.

- Regardless of whether or not it is selected as the system clock, confirm that the OSCOVFSR.MOOVF flag has been set to 1 before executing a WAIT instruction to place the chip to software standby or deep software standby after the setting to operate the main clock oscillator by the MOSTP bit.
- When a transition to software standby or deep software standby is to follow the setting to stop the main clock oscillator, confirm that the OSCOVFSR.MOOVF flag has been set to 0 before executing the WAIT instruction.

Writing of 1 to the MOSTP bit (stopping the main clock oscillator) is prohibited in either of the following cases:

- The main clock oscillator is selected by the clock source select bits in system clock control register 3 (SCKCR3.CKSEL[2:0]).
- 0 (PLL operation) is selected by the PLL stop control bit in PLL control register 2 (PLLCR2.PPLEN), and the main clock oscillator is selected by the PLL clock source select bit in the PLL control register (PLLCR.PLLSRCSEL).
- 0 (PPLL operation) is selected by the PPLL stop control bit in PPLL control register 2 (PPLLCR2.PPLEN), and the main clock oscillator is selected by the PLL clock source select bit in the PLL control register (PLLCR.PLLSRCSEL).

9.2.9 Sub-Clock Oscillator Control Register (SOSCCR)

Address(es): 0008 0033h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	SOSTP
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	SOSTP	Sub-Clock Oscillator Stop	0: Sub-clock oscillator is operating. 1: Sub-clock oscillator is stopped.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

SOSTP Bit (Sub-Clock Oscillator Stop)

This bit runs or stops the sub-clock oscillator.

The SOSTP bit and the sub-clock control bit in the RTC control register 3 (RCR3.RTCEN) controls whether to operate or stop the sub-clock oscillator. If one of these bits is set so as to enable the operation, the sub-clock oscillator runs.

When changing the value of the SOSTP bit or RCR3.RTCEN bit, execute subsequent instructions after reading the bit and checking that its value has actually been updated (refer to (2) Notes on Writing to I/O Registers, in section 5, I/O Registers).

When the sub-clock is to be used as the source to drive counting by the realtime clock, use the RCR3.RTCEN and SOSTP bits. When the sub-clock is not to be used as the clock source for counting by the realtime clock but is to be used as the system clock, use the SOSCCR.SOSTP bit.

When the SOSTP bit is set to operate the sub-clock, be sure to set the sub-clock oscillator wait control register (SOSCWTCR) beforehand. Furthermore, after selecting sub-clock operation, only start using the sub-clock after confirming that the OSCOVFSR.SOOVF flag has been set to 1.

A fixed time for stabilization is required for oscillation to become stable after selecting sub-clock operation with the SOSTP bit. Furthermore, a fixed time is required for oscillation to actually stop after the setting to stop operation. Accordingly, take note of the following when starting and stopping operation with the SOSTP bit.

- Sub-clock operation can be selected with the SOSTP bit regardless of the setting of the OSCOVFSR.SOOVF flag. However, the time until deactivation of the sub-clock is completed (the time until the OSCOVFSR.SOOVF flag is set to 0 after the setting to stop operation) means that writing to the SOSCCR register takes longer than the setting to operate the sub-clock.
- The sub-clock can be stopped by the SOSTP bit regardless of the setting of the OSCOVFSR.SOOVF flag. However, the wait for oscillation to become stable, i.e. the oscillation settling time (the time until the OSCOVFSR.SOOVF flag is set to 1 after the setting to operate the sub-clock), means that writing to the SOSCCR register takes longer than the setting to stop operation.
- Regardless of whether or not it is selected as the system clock, confirm that the OSCOVFSR.SOOVF flag has been set to 1 before executing a WAIT instruction to place the chip on software standby or deep software standby after the setting to operate the sub-clock oscillator by the SOSTP bit.
- When a transition to software standby or deep software standby is to follow the setting to stop the sub-clock oscillator, confirm that the OSCOVFSR.SOOVF flag has been set to 0 after the setting to stop the sub-clock oscillator and before executing the WAIT instruction.

Writing of 1 to the SOSTP bit (stopping the sub-clock oscillator) is prohibited while the sub-clock oscillator is selected by the clock source select bits in system clock control register 3 (SCKCR3.CKSEL[2:0]).

9.2.10 Low-Speed On-Chip Oscillator Control Register (LOCOCR)

Address(es): 0008 0034h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	LCSTP
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	LCSTP	LOCO Stop	0: LOCO is operating. 1: LOCO is stopped.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

LCSTP Bit (LOCO Stop)

This bit runs or stops the LOCO.

After the setting of the LCSTP bit has been changed so that the LOCO operates, only start using the LOCO after the LOCO clock oscillation stabilization waiting time (tLOCOWT) has elapsed.

That is, a fixed time for stabilization of oscillation is required for oscillation to become stable after setting LOCO operation with the LCSTP bit. A fixed time is also required for oscillation to stop after the setting to stop the oscillator. Accordingly, take note of the following limitations when starting and stopping the oscillator.

- When restarting the LOCO after it has been stopped, allow at least five cycles of the LOCO as an interval over which it is still stopped.
- Ensure that oscillation by the LOCO is stable when making the setting to stop the LOCO.
- Regardless of whether or not it is selected as the system clock, ensure that oscillation by the LOCO is stable before executing a WAIT instruction to place the chip on software standby or deep software standby.
- When a transition to software standby or deep software standby is to follow the setting to stop the LOCO, wait for at least three cycles of the LOCO after the setting to stop the LOCO and before executing the WAIT instruction.

Writing of 1 to the LCSTP bit (stopping the LOCO) is prohibited while the LOCO is selected by the clock source select bits in system clock control register 3 (SCKCR3.CKSEL[2:0]).

Writing of 1 to the LCSTP bit (stopping the LOCO) is prohibited if detection of oscillation stopping is enabled by the oscillation stop detection enable bit in the oscillation stop detection control register (OSTDCR.OSTDE).

Since the LOCO clock is used to measure the waiting time for other oscillators, the LOCO clock oscillates while the waiting time for other oscillators is being measured, regardless of the setting of LCSTP bit. Therefore, the LOCO clock may be unintentionally supplied even if the LCSTP bit is set to be stopped.

9.2.11 IWDT-Dedicated On-Chip Oscillator Control Register (ILOCOCR)

Address(es): 0008 0035h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	ILCSTP
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Bit Name	Description	R/W
b0	ILCSTP	IWDT-Dedicated On-Chip Oscillator Stop	0: IWDT-dedicated on-chip oscillator is operating. 1: IWDT-dedicated on-chip oscillator is stopped.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

When the IWDT start mode select bit in the option function select register 0 (OFS0.IWDTSTRT) is 0 (IWDT operating), the setting of this register is invalid; it is valid only when the OFS0.IWDTSTRT bit is set to 1 (IWDT stopped). The ILCSTP bit cannot be changed from 0 (IWDT-dedicated on-chip oscillator operating) to 1 (IWDT-dedicated on-chip oscillator stopped) while ILOCOCR is valid.

ILCSTP Bit (IWDT-Dedicated On-Chip Oscillator Stop)

This bit runs or stops the IWDT-dedicated on-chip oscillator.

After the setting of the ILCSTP bit has been changed to make the IWDT-dedicated on-chip oscillator run, confirm that the OSCOVFSR.ILCOVF flag has been set to 1 before starting to use the oscillator.

When a transition to software standby or deep software standby mode is to follow the setting to start the IWDT-dedicated on-chip oscillator, confirm that the OSCOVFSR.ILCOVF flag has been set to 1 before executing the WAIT instruction.

9.2.12 High-Speed On-Chip Oscillator Control Register (HOCOCR)

Address(es): 0008 0036h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	HCSTP
0	0	0	0	0	0	0	0/1**1

Bit	Symbol	Bit Name	Description	R/W
b0	HCSTP	HOCO Stop	0: HOCO is operating. 1: HOCO is stopped.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. The HCSTP bit value after a reset is 0 when the OFS1.HOCOEN bit is 0. It is 1 when the OFS1.HOCOEN bit is 1.

HCSTP Bit (HOCO Stop)

This bit runs or stops the HOCO.

After the setting of the HCSTP bit has been changed to make the HOCO run, confirm that the OSCOVFSR.HCOVF flag has been set to 1 before starting to use the oscillator.

A fixed time for stabilization is required for oscillation to become stable after setting HOCO operation with the HCSTP bit. Furthermore, a fixed time is required for oscillation to actually stop after the setting to stop operation. Accordingly, take note of the following when starting and stopping operation with the HCSTP bit.

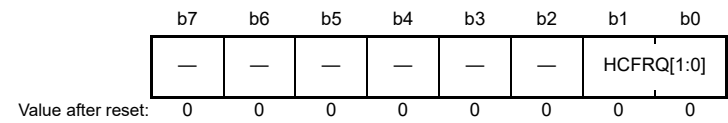
- Setting HOCO operation with the HCSTP bit is possible regardless of the setting of the OSCOVFSR.HCOVF flag. However, the time until deactivation of the HOCO is completed (the time until the OSCOVFSR.HCOVF flag is set to 0 after the setting to stop HOCO operation) means that writing to the HOCOCR register takes longer than the setting to operate the HOCO.
- The HOCO can be stopped by the HCSTP bit regardless of the setting of the OSCOVFSR.HCOVF flag. However, the wait for oscillation to become stable, i.e. the oscillation settling time (the time until the OSCOVFSR.HCOVF flag is set to 1 after the setting to operate the HOCO), means that writing to the HOCOCR register takes longer than the setting to stop HOCO operation.
- Regardless of whether or not it is selected as the system clock, confirm that the OSCOVFSR.HCOVF flag has been set to 1 before executing a WAIT instruction to place the chip on software standby or deep software standby after selecting HOCO operation with the HCSTP bit.
- When a transition to software standby or deep software standby is to follow the setting to stop the HOCO, confirm that the OSCOVFSR.HCOVF flag has been set to 0 after the setting to stop the HOCO and before executing the WAIT instruction.

Writing that causes setting of the HCSTP bit to 1 (stopping the HOCO) is prohibited while any of the following is selected: the HOCO is selected in the clock source select bits (SCKCR3.CKSEL[2:0]) in system clock control register 3; the HOCO is selected as the clock source for the PLL in the PLLCR.PLLSRCSEL bit and the PLL is selected in the PSCKCR3.CKSEL[2:0] bits; or the HOCO is selected as the clock source for the PLL by the PLLCR.PLLSRCSEL bit and the setting of the PPLLCR2.PPLEN bit is 0 (PLL is operating).

Writing of 0 to the HCSTP bit (making the HOCO operate) is prohibited when the setting of the operating power control mode select bits in the operating power control register (OPCCR.OPCM[2:0]) is for low-speed operating mode 2.

9.2.13 High-Speed On-Chip Oscillator Control Register 2 (HOCOCR2)

Address(es): 0008 0037h



Bit	Symbol	Bit Name	Description	R/W
b1, b0	HCFRQ[1:0]	HOCO Frequency Setting	b1 b0 0 0: 16 MHz 0 1: 18 MHz 1 0: 20 MHz Settings other than above are prohibited.	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Writing to the HOCOCR2 register is prohibited when the HOCOCR.HCSTP bit is 0 (making the HOCO run).

9.2.14 Oscillation Stabilization Flag Register (OSCOVFSR)

Address(es): 0008 003Ch

b7	b6	b5	b4	b3	b2	b1	b0
—	—	PPLOV F	ILCOV F	HCOVF	PLOVF	SOOVF	MOOV F
Value after reset:	0	0	0	0/1*1	0/1*2	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	MOOVF	Main Clock Oscillation Stabilization Flag	0: MOSTP = 1 (stopping the main clock oscillator) or oscillation of the main clock has not yet become stable.*3, *5 1: Oscillation of the main clock is stable so the clock is available for use as the system clock.*4	R
b1	SOOVF	Sub-Clock Oscillation Stabilization Flag	0: SOSTP = 1 (stopping the sub-clock oscillator) or oscillation of the sub-clock has not yet become stable.*5 1: Oscillation of the sub-clock is stable so the clock is available for use as the system clock.*4	R
b2	PLOVF	PLL Clock Oscillation Stabilization Flag	0: The PLL clock is stopped or oscillation of the PLL clock has not yet become stable. 1: Oscillation of the PLL clock is stable so the clock is available for use as the system clock.	R
b3	HCOVF*2	HOCO Clock Oscillation Stabilization Flag	0: The HOCO clock is stopped or oscillation of the HOCO clock has not yet become stable. 1: Oscillation of the HOCO clock is stable so the clock is available for use as the system clock.	R
b4	ILCOVF*1	IWDT-Dedicated Clock Oscillation Stabilization Flag	0: The IWDT-dedicated on-chip oscillator is stopped or oscillation of the IWDT-dedicated on-chip oscillator has not yet become stable. 1: Oscillation of the IWDT-dedicated on-chip oscillator is stable.	R
b5	PPLOVF	PPLL Clock Oscillation Stabilization Flag	0: The PPLL clock is stopped or oscillation of the PPLL clock has not yet become stable. 1: Oscillation of the PPLL clock is stable.	R
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. The ILCOVF flag value after a reset is 1 when the OFS0.IWDTSTRT bit is 0. It is 0 when the OFS0.IWDTSTRT bit is 1.

Note 2. The HCOVF flag value after a reset is 1 when the OFS1.HOCOEN bit is 0. It is 0 when the OFS1.HOCOEN bit is 1.

Note 3. The MOOVF flag does not reflect the control of the main clock oscillator by the MOFCR.MOFXIN bit. Accordingly, if the MOSTP bit is set to 1 while the MOFXIN bit is 1, the main clock oscillator continues to oscillate but the setting of the MOOVF flag becomes 0.

Note 4. The SOOVF flag does not reflect the control of the sub-clock oscillator by the RCR3.RTCEN bit. Accordingly, if the SOSTP bit is set to 1 while the RTCEN bit is 1, the sub-clock oscillator continues to oscillate but the setting of the SOOVF flag becomes 0.

Note 5. If the value set in the wait control register of the main clock oscillator and sub-clock oscillator is not sufficient for the given oscillation stabilization time, the oscillation stabilization flag is set to 1 and supply of the clock signal to the internal circuits starts before oscillation is stable. This may cause malfunction of this MCU, so ensure that the setting of the wait control register is at least the oscillation settling time for the oscillator considering the maximum frequency of the LOCO clock.

OSCOVFSR contains flags to indicate the states of operation of the counters within the oscillation stabilization wait circuits for the individual oscillators.

The counters measure the waiting times until each oscillator output clock is supplied to the internal circuits after oscillation starts, and an overflow of a counter indicates the start of clock supply from the corresponding oscillator to the internal circuits.

MOOVF Flag (Main Clock Oscillation Stabilization Flag)

This flag indicates the state of operation of the counter that measures the waiting time for the main clock oscillator.

[Setting condition]

- After the main clock oscillator has stopped and the MOSCCR.MOSTP bit is set to 0, the number of LOCO clock cycles corresponding to the setting of the MOSCWTCR register being counted and supply of the main clock within the MCU starting.

[Clearing condition]

- After the main clock oscillator has started to operate and the MOSCCR.MOSTP bit has been set to 1, deactivation of the main clock oscillator being completed.

SOOVF Flag (Sub-Clock Oscillation Stabilization Flag)

This flag indicates the state of operation of the counter that measures the waiting time for the sub-clock oscillator.

[Setting condition]

- After the sub-clock oscillator has stopped and the SOSCCR.SOSTP bit is set to 0, the number of LOCO clock cycles corresponding to the setting of the SOSCWTCR register being counted and supply of the sub-clock within the MCU starting.

[Clearing condition]

- After the sub-clock oscillator has started to operate and the SOSCCR.SOSTP bit has been set to 1, deactivation of the sub-clock oscillator being completed.

PLOVF Flag (PLL Clock Oscillation Stabilization Flag)

This flag indicates the state of operation of the counter that measures the waiting time for the PLL.

[Setting condition]

- After the PLL has stopped and the PLLCR2.PLEN bit is set to 0, 62 cycles of the LOCO clock being counted and supply of the PLL clock within the MCU starting.

If oscillation by the PLL clock source selected by the PLLCR.PLLSRCSEL bit is not stable when the PLEN bit is set to 0, counting of LOCO clock cycles proceeds after the oscillation of the PLL clock source has been stabilized.

[Clearing condition]

- After the PLL has started to operate and the PLLCR2.PLEN bit has been set to 1, deactivation of the PLL being completed.

HCOVF Flag (HOCO Clock Oscillation Stabilization Flag)

This flag indicates the state of operation of the counter that measures the waiting time for the high-speed on-chip oscillator.

[Setting condition]

- After the high-speed on-chip oscillator has stopped and the HOCOCR.HCSTP bit is set to 0, 25 cycles of the LOCO clock being counted and supply of the HOCO clock within the MCU starting.

[Clearing condition]

- After the high-speed on-chip oscillator has started to operate and the HOCOCR.HCSTP bit has been set to 1, deactivation of the high-speed on-chip oscillator being completed.

ILCOVF Flag (IWDT-Dedicated Clock Oscillation Stabilization Flag)

This flag indicates the state of operation of the counter that measures the waiting time for the IWDT-dedicated on-chip oscillator.

[Setting condition]

- After the IWDT-dedicated on-chip oscillator has stopped and the ILOCOCR.ILCSTP bit is set to 0, 34 cycles of the LOCO clock being counted and supply of the IWDT-dedicated clock within the MCU starting.

PPLOVF Flag (PPLL Clock Oscillation Stabilization Flag)

This flag indicates the state of operation of the counter that measures the waiting time for the PPLL.

[Setting condition]

- After the PPLL has stopped and the PPLLCR2.PPLEN bit is set to 0, 62 cycles of the LOCO clock being counted and supply of the PPLL clock within the MCU starting.

If oscillation by the PPLL clock source selected by the PLLCR.PLLSRCSEL bit is not stable when the PPLEN bit is set to 0, counting of LOCO clock cycles proceeds after the oscillation of the PPLL clock source has been stabilized.

[Clearing condition]

- After the PPLL has started to operate and the PPLLCR2.PPLEN bit has been set to 1, deactivation of the PPLL being completed.

9.2.15 Oscillation Stop Detection Control Register (OSTDCR)

Address(es): 0008 0040h

	b7	b6	b5	b4	b3	b2	b1	b0
	OSTDE	—	—	—	—	—	—	OSTDIE
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	OSTDIE	Oscillation Stop Detection Interrupt Enable	0: The oscillation stop detection interrupt is disabled. Oscillation stop detection is not notified to the POE. 1: The oscillation stop detection interrupt is enabled. Oscillation stop detection is notified to the POE.	R/W
b6 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b7	OSTDE	Oscillation Stop Detection Function Enable	0: Oscillation stop detection function is disabled. 1: Oscillation stop detection function is enabled.	R/W

OSTDCR register is used to enable the oscillation stop detection function for the main clock oscillator and conveying of interrupts in response.

OSTDIE Bit (Oscillation Stop Detection Interrupt Enable)

If the oscillation-stop detection flag in the oscillation-stop detection status register (OSTDSR.OSTDF) requires clearing, do this after clearing the OSTDIE bit to 0. Wait for at least two cycles of PCLKB before again setting the OSTDIE bit to 1. According to the number of cycles for access to read out a given I/O register, waiting time longer than two cycles of PCLKB may have to be secured.

OSTDE Bit (Oscillation Stop Detection Function Enable)

This bit enables or disables the oscillation stop detection function.

When the OSTDE bit is 1 (oscillation stop detection function enabled), the LOCO stop bit (LOCOCR.LCSTP) is set to 0 and the LOCO operation is started. The LOCO cannot be stopped while the oscillation stop detection function is enabled; writing 1 to the LOCOCR.LCSTP bit (LOCO stopped) is invalid.

When the oscillation stop detection flag in the oscillation stop detection status register (OSTDSR.OSTDF) is 1 (main clock oscillation stop detected), writing 0 to the OSTDE bit is invalid.

When the OSTDE bit is 1, a transition cannot be made to software standby mode or deep software standby mode. To make a transition to software standby mode or deep software standby mode, execute the WAIT instruction with the OSTDE bit being 0.

To check the oscillation stop detection flag in the oscillation stop detection status register (OSTDSR.OSTDF), wait for at least three cycles of ICLK after the OSTDE bit has been set to 1 (oscillation stop detection enabled).

9.2.16 Oscillation Stop Detection Status Register (OSTDSR)

Address(es): 0008 0041h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	OSTDF
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	OSTDF	Oscillation Stop Detection Flag	0: Stopping of the main clock oscillator has not been detected. 1: Stopping of the main clock oscillator has been detected.	R/(W) *1
b7 to b1	—	Reserved	These bits are read as 0 and cannot be modified.	R

Note 1. This bit can only be set to 0.

The OSTDSR register indicates the detection of stopping of the main clock oscillator.

OSTDF Flag (Oscillation Stop Detection Flag)

This bit is a flag to indicate the main clock status. When the OSTDF flag is 1, it indicates that the main clock oscillation stop has been detected.

Once the main clock oscillation stop is detected, the OSTDF flag is not set to 0 even though the main clock oscillation is restarted. The OSTDF flag is set to 0 by reading 1 from the bit and then writing 0. At least 3 ICLK cycles of wait time is necessary between writing 0 to OSTDF and reading OSTDF as 0. If the OSTDF flag is set to 0 from 1 while the main clock oscillation is stopped, the OSTDF flag becomes 0 and then returns to 1.

The OSTDF flag cannot be modified to 0 while the main clock oscillator or PLL is selected by the clock source select bits in system clock control register 3 (SCKCR3.CKSEL[2:0]) (010b or 100b). The OSTDF flag should be set to 0 after switching the clock source to other sources than the main clock oscillator and PLL.

[Setting condition]

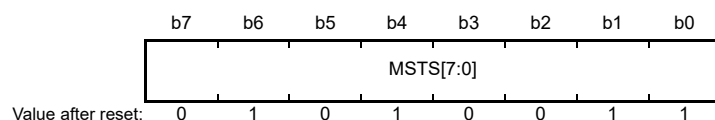
- The main clock oscillation is stopped with the OSTDCR.OSTDE being 1 (oscillation stop detection function enabled).

[Clearing condition]

- When 1 is read and then 0 is written when the value of the SCKCR3.CKSEL[2:0] bits are neither 010b or 100b and the PLLCR.PLLSRCSEL bit is set to a value other than 0.

9.2.17 Main Clock Oscillator Wait Control Register (MOSCWTCR)

Address(es): 0008 00A2h



MOSCWTCR is used to control the waiting time until output of the signal from the main clock oscillator to the internal circuits starts. The oscillation stabilization wait circuit for the main clock oscillator measures the waiting time by counting the number of LOCO clock cycles corresponding to the setting of the MOSCWTCR register.

The oscillation stabilization wait circuit measures the waiting time and controls the clock supply within the MCU. When the main clock oscillator starts by setting the MOSCCR.MOSTP bit, the oscillation stabilization wait circuit starts counting the waiting time with the LOCO clock. The clock supply within the MCU is disabled over the period until counting of the set number of cycles is completed. After counting is completed, supply of the clock signal within the MCU starts and the OSCOVFSR.MOOVF flag is set to 1.

Counting of LOCO clock cycles by the oscillation stabilization wait circuit proceeds regardless of the setting of the LOCOCR.LCSTP bit. Hardware automatically controls running and stopping of the LOCO clock for measurement of the waiting time.

Values can only be written to MOSCWTCR while the MOSCCR.MOSTP bit is 1 or the OSCOVFSR.MOOVF flag is 1; do not attempt writing to MOSCWTCR if neither is the case.

The waiting time is not required when an external clock signal is input for the main clock oscillator. Set the MSTS[7:0] bits to 00h.

The value of the MSTS[7:0] bits required for correspondence with the waiting time required to secure stable oscillation by the main clock oscillator is obtained by using the maximum frequency for fLOCO in the formula below.

$$\text{MSTS}[7:0] > (\text{tMAINOSC} \times (\text{fLOCO_max}) + 16)/32$$

(tMAINOSC: main clock oscillation stabilization time; fLOCO_max: maximum frequency for fLOCO)

If tMAINOSC is 1 ms and fLOCO_max is 264 kHz (the period is 1/3.78 μs), the formula gives MSTS[7:0] > (1 ms × (264 kHz) + 16)/32 = 8.75, so set the MSTS[7:0] bits to 9.

Waiting time:

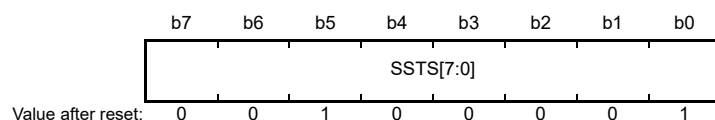
When LOCO is at its highest frequency: $(9 \times 32 - 16) \times (1/264 \text{ kHz} = 3.78 \mu\text{s}) = 1.028 \text{ ms}$

When LOCO is at its normal frequency: $(9 \times 32 + 3) \times (1/240 \text{ kHz} = 4.18 \mu\text{s}) = 1.216 \text{ ms}$

When LOCO is at its lowest frequency: $(9 \times 32 + 10) \times (1/216 \text{ kHz} = 4.63 \mu\text{s}) = 1.380 \text{ ms}$

9.2.18 Sub-Clock Oscillator Wait Control Register (SOSCWTCR)

Address(es): 0008 00A3h



SOSCWTCR is used to control the waiting time until output of the signal from the sub-clock oscillator to the internal circuits starts. The oscillation stabilization wait circuit for the sub-clock oscillator measures the waiting time by counting the number of LOCO clock cycles corresponding to the setting of the SOSCWTCR register.

The oscillation stabilization wait circuit measures the waiting time and controls the clock supply within the MCU. When the sub-clock oscillator starts by setting the SOSCCR.SOSTP bit, the oscillation stabilization wait circuit starts counting the waiting time with the LOCO clock. The clock supply within the MCU is disabled over the period until counting of the set number of cycles is completed. After counting is completed, supply of the clock signal within the MCU starts and the OSCOVFSR.SOOVF flag is set to 1.

Counting of LOCO clock cycles by the oscillation stabilization wait circuit proceeds regardless of the setting of the LOCOCR.LCSTP bit. Hardware automatically controls running and stopping of the LOCO clock for measurement of the waiting time.

Values can only be written to SOSCWTCR while the SOSCCR.SOSTP bit is 1 or the OSCOVFSR.SOOVF flag is 1; do not attempt writing to SOSCWTCR if neither is the case.

The value of the SSTS[7:0] bits required for correspondence with the expected time to secure settling of oscillation by the sub-clock oscillator is obtained by using the maximum frequency for fLOCO in the formula below.

$$SSTS[7:0] > (t_{SUBOSC} \times (f_{LOCO_max}) + 16) / 16384$$

(tSUBOSC: sub-clock oscillation stabilization time; fLOCO_max: maximum frequency for fLOCO)

If tSUBOSC is 2 s and fLOCO is 264 kHz (the period is 1/3.78 μs), the formula gives SSTS[7:0] > (2 s × (264 kHz) + 16)/16384 = 32.22, so set the SSTS[7:0] bits to 33.

Waiting time:

When LOCO is at its highest frequency: $(33 \times 16384 - 16) \times (1/264 \text{ kHz} = 3.78 \text{ μs}) = 2.044 \text{ s}$

When LOCO is at its normal frequency: $(33 \times 16384 + 3) \times (1/240 \text{ kHz} = 4.18 \text{ μs}) = 2.260 \text{ s}$

When LOCO is at its lowest frequency: $(33 \times 16384 + 10) \times (1/216 \text{ kHz} = 4.63 \text{ μs}) = 2.503 \text{ s}$

9.2.19 Main Clock Oscillator Forced Oscillation Control Register (MOFCR)

Address(es): 0008 C293h

b7	b6	b5	b4	b3	b2	b1	b0
—	MOSEL	MODRV2[1:0]	—	—	—	—	MOFXIN
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	MOFXIN	Main Clock Oscillator Forced Oscillation	0: Oscillator is not controlled by this bit. 1: The main clock oscillator is forcedly oscillated.	R/W
b3 to b1	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b5, b4	MODRV2[1:0]	Main Clock Oscillator Driving Ability 2 Switching	b5 b4 0 0: 20.1 to 24 MHz 0 1: 16.1 to 20 MHz 1 0: 8.1 to 16 MHz 1 1: 8 MHz	R/W
b6	MOSEL	Main Clock Oscillator Switching	0: Resonator 1: External clock input	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

As well as selecting and de-selecting forcible starting of the main clock oscillator, the setting of the MOFCR also selects the driving ability, and the oscillator or an external clock signal.

MOFXIN Bit (Main Clock Oscillator Forced Oscillation)

This bit controls forced oscillation of the main clock oscillator. Although transitions to software standby or deep software standby normally stop oscillation of the main clock oscillator, setting this bit to 1 forces oscillation by the main clock oscillator so that it can still be used as the clock source for the RTC in either standby mode.

When changing the value of the MOSCCR.MOSTP bit or MOFXIN bit, execute subsequent instructions after reading the bit and checking that its value has actually been updated (refer to (2) Notes on Writing to I/O Registers, in section 5, I/O Registers).

MODRV2[1:0] Bits (Main Clock Oscillator Driving Ability 2 Switching)

These bits switch the driving ability of the main clock oscillator.

Specify the driving ability according to the frequency of a crystal connected to the main clock oscillator.

The frequency ranges specified in the bit description of the MODRV2[1:0] bits are the reference values of the crystal with capacitive load of 8 pF. A setting value may not fit within the frequency range depending on a crystal. Use values recommended by the resonator manufacturer.

In case of a ceramic resonator, it may be better to select lower frequency range than the frequency of the resonator (for example, specify 10b instead of 01b when a ceramic resonator with the frequency range from 16.1 to 20 MHz is used). Use values recommended by the resonator manufacturer.

MOSEL Bit (Main Clock Oscillator Switching)

This bit switches the source for the main clock oscillator.

9.2.20 High-Speed On-Chip Oscillator Power Supply Control Register (HOCOPCR)

Address(es): 0008 C294h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	HOCO PCNT
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	HOCOPCNT	High-Speed On-Chip Oscillator Power Supply Control	0: Turns the power supply of the HOCO on. 1: Turns the power supply of the HOCO off.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

HOCOPCNT Bit (High-Speed On-Chip Oscillator Power Supply Control)

This bit controls the power supply for the HOCO.

When this bit is set to 0, the power supply of the HOCO is turned on, enabling oscillation.

When this bit is set to 1, the power supply of the HOCO is turned off, reducing power consumption.

When setting the HOCOPCNT bit to 1, set the HOCO stop bit in the high-speed on-chip oscillator control register (HOCOCR.HCSTP) to 1 (HOCO stopped) beforehand.

After the HOCOPCNT bit is changed from 1 to 0, oscillation settling time is required before the HOCOCR.HCSTP bit is set to 0. For details, refer to [section 63, Electrical Characteristics](#).

Do not change the value of the HOCOPCNT bit in the following cases:

- When the HOCO is selected as the clock source by the clock source select bits in system clock control register 3 (SCKCR3.CKSEL[2:0]).
- When the setting of the operating power control mode select bits in the operating power control register (OPCCR.OPCM[2:0]) is for low-speed operating mode 1 or 2.

9.2.21 CLKOUT Output Control Register (CKOCR)

Address(es): 0008 003Eh

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	CKOSTP	CKODIV[2:0]			—	CKOSEL[2:0]			—	—	—	—	—	—	—	—
Value after reset:	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b10 to b8	CKOSEL[2:0]	CLKOUT Output Source Select	b10 b8 0 0 0: LOCO 0 0 1: HOCO 0 1 0: Main clock oscillator 0 1 1: Sub-clock oscillator 1 0 0: PLL circuit 1 1 0: PPLL circuit Settings other than above are prohibited.	R/W
b11	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b14 to b12	CKODIV[2:0]	CLKOUT Output Divisor Select	b14 b12 0 0 0: $\times 1/1$ 0 0 1: $\times 1/2$ 0 1 0: $\times 1/4$ 0 1 1: $\times 1/8$ 1 0 0: $\times 1/16$ Settings other than above are prohibited.	R/W
b15	CKOSTP	CLKOUT Output Stop Control	0: CLKOUT pin output enabled*1 1: CLKOUT pin output stopped (fixed to the low level)	R/W

Note 1. Setting of the pin function control register and port mode register for the corresponding pin is also required.

The CKOCR register sets the clock signal for output from the CLKOUT pin.

CKOSEL[2:0] Bits (CLKOUT Output Source Select)

The CKOSEL[2:0] Bits select the clock source for output from the CLKOUT pin from among the LOCO clock, HOCO clock, main clock, sub-clock, PLL clock, and PPLL clock.

To change the clock source, start by setting the CKOSTP bit to 1. Selecting a clock source that is stopped is prohibited.

CKODIV[2:0] Bits (CLKOUT Output Divisor Select)

The CKODIV[2:0] bits select the divisor for the clock selected in the CKOSEL[2:0] bits.

To change the ratio, start by setting the CKOSTP bit to 1.

CKOSTP Bit (CLKOUT Output Stop Control)

The CKOSTP bit controls the output from the CLKOUT pin.

Setting the bit to 0 leads to output of the selected clock. Setting the bit to 1 drives the output to the low level.

Changing the value of the CKOSTP bit while the clock is oscillating may generate a glitch in the output.

9.2.22 Specific-Use Clock Control Register (PACKCR)

Address(es): 0008 0044h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	UPLLS EL	—	—	—	—	—	—	—	OUTCK SEL	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Bit Name	Description	R/W
b0	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b3 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b4	OUTCKSEL	Ethernet-PHY External Clock (CLKOUT25M) Source Select	0: PLL circuit 1: PPLL circuit	R/W
b11 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b12	UPLLSEL	USB Clock (UCLK) Source Select	0: USB frequency-divided clock (UDIVCLK) 1: PPLL frequency-divided clock (PPLLDIVCLK)	R/W
b15 to b13	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The PACKCR register selects the clock source for the Ethernet-PHY external clock (CLKOUT25M), and USB clock (UCLK).

OUTCKSEL Bit (Ethernet-PHY External Clock (CLKOUT25M) Source Select)

The OUTCKSEL bit selects the source of the Ethernet-PHY external clock (CLKOUT25M) as either the PLL or PPLL circuit.

Writing to this bit while the PLL or PPLL is oscillating is prohibited.

Also, only set this bit once after release from the reset state.

UPLLSEL Bit (USB Clock (UCLK) Source Select)

The UPLLSEL bit selects the source of the USB clock (UCLK) as either the USB frequency-divided clock (UDIVCLK) or PPLL frequency-divided clock (PPLLDIVCLK).

To switch the bit from 0 to 1, do so while oscillation by the PPLL circuit is stopped. Also, only set this bit once after a release from the reset state while the supply of the clock signal to the USB module is stopped (MSTPCRB.MSTPB19 bit = 1).

9.2.23 PPLL Control Register (PPLLCR)

Address(es): 0008 0048h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	PPLSTC[5:0]					—	—	—	—	—	—	—	PPLIDIV[1:0]	
0	0	0	1	1	1	0	1	0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W																																																																																										
b1, b0	PPLIDIV[1:0]	PPLL Input Pulse Frequency Division Ratio Select	b1 b0 0 0: ×1/1 0 1: ×1/2 1 0: ×1/3 1 1: Setting prohibited	R/W																																																																																										
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W																																																																																										
b13 to b8	PPLSTC[5:0]	PPLL Frequency Multiplier Setting	<table><tr><td>b13</td><td>b8</td><td>b13</td><td>b8</td><td>b13</td><td>b8</td></tr><tr><td>0 1 0 0 1 1</td><td>×10.0</td><td>1 0 0 0 0 1</td><td>×17.0</td><td>1 0 1 1 1 1</td><td>×24.0</td></tr><tr><td>0 1 0 1 0 0</td><td>×10.5</td><td>1 0 0 0 1 0</td><td>×17.5</td><td>1 1 0 0 0 0</td><td>×24.5</td></tr><tr><td>0 1 0 1 0 1</td><td>×11.0</td><td>1 0 0 0 1 1</td><td>×18.0</td><td>1 1 0 0 0 1</td><td>×25.0</td></tr><tr><td>0 1 0 1 1 0</td><td>×11.5</td><td>1 0 0 1 0 0</td><td>×18.5</td><td>1 1 0 0 1 0</td><td>×25.5</td></tr><tr><td>0 1 0 1 1 1</td><td>×12.0</td><td>1 0 0 1 0 1</td><td>×19.0</td><td>1 1 0 0 1 1</td><td>×26.0</td></tr><tr><td>0 1 1 0 0 0</td><td>×12.5</td><td>1 0 0 1 1 0</td><td>×19.5</td><td>1 1 0 1 0 0</td><td>×26.5</td></tr><tr><td>0 1 1 0 0 1</td><td>×13.0</td><td>1 0 0 1 1 1</td><td>×20.0</td><td>1 1 0 1 0 1</td><td>×27.0</td></tr><tr><td>0 1 1 0 1 0</td><td>×13.5</td><td>1 0 1 0 0 0</td><td>×20.5</td><td>1 1 0 1 1 0</td><td>×27.5</td></tr><tr><td>0 1 1 0 1 1</td><td>×14.0</td><td>1 0 1 0 0 1</td><td>×21.0</td><td>1 1 0 1 1 1</td><td>×28.0</td></tr><tr><td>0 1 1 1 0 0</td><td>×14.5</td><td>1 0 1 0 1 0</td><td>×21.5</td><td>1 1 1 0 0 0</td><td>×28.5</td></tr><tr><td>0 1 1 1 0 1</td><td>×15.0</td><td>1 0 1 0 1 1</td><td>×22.0</td><td>1 1 1 0 0 1</td><td>×29.0</td></tr><tr><td>0 1 1 1 1 0</td><td>×15.5</td><td>1 0 1 1 0 0</td><td>×22.5</td><td>1 1 1 0 1 0</td><td>×29.5</td></tr><tr><td>0 1 1 1 1 1</td><td>×16.0</td><td>1 0 1 1 0 1</td><td>×23.0</td><td>1 1 1 0 1 1</td><td>×30.0</td></tr><tr><td>1 0 0 0 0 0</td><td>×16.5</td><td>1 0 1 1 1 0</td><td>×23.5</td><td></td><td></td></tr></table> Settings other than above are prohibited.	b13	b8	b13	b8	b13	b8	0 1 0 0 1 1	×10.0	1 0 0 0 0 1	×17.0	1 0 1 1 1 1	×24.0	0 1 0 1 0 0	×10.5	1 0 0 0 1 0	×17.5	1 1 0 0 0 0	×24.5	0 1 0 1 0 1	×11.0	1 0 0 0 1 1	×18.0	1 1 0 0 0 1	×25.0	0 1 0 1 1 0	×11.5	1 0 0 1 0 0	×18.5	1 1 0 0 1 0	×25.5	0 1 0 1 1 1	×12.0	1 0 0 1 0 1	×19.0	1 1 0 0 1 1	×26.0	0 1 1 0 0 0	×12.5	1 0 0 1 1 0	×19.5	1 1 0 1 0 0	×26.5	0 1 1 0 0 1	×13.0	1 0 0 1 1 1	×20.0	1 1 0 1 0 1	×27.0	0 1 1 0 1 0	×13.5	1 0 1 0 0 0	×20.5	1 1 0 1 1 0	×27.5	0 1 1 0 1 1	×14.0	1 0 1 0 0 1	×21.0	1 1 0 1 1 1	×28.0	0 1 1 1 0 0	×14.5	1 0 1 0 1 0	×21.5	1 1 1 0 0 0	×28.5	0 1 1 1 0 1	×15.0	1 0 1 0 1 1	×22.0	1 1 1 0 0 1	×29.0	0 1 1 1 1 0	×15.5	1 0 1 1 0 0	×22.5	1 1 1 0 1 0	×29.5	0 1 1 1 1 1	×16.0	1 0 1 1 0 1	×23.0	1 1 1 0 1 1	×30.0	1 0 0 0 0 0	×16.5	1 0 1 1 1 0	×23.5			R/W
b13	b8	b13	b8	b13	b8																																																																																									
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1 0 0 0 0 0	×16.5	1 0 1 1 1 0	×23.5																																																																																											
b15, b14	—	Reserved	These bits are read as 0. The write value should be 0.	R/W																																																																																										

Writing to the PPLLCR register while the PPLLCR2.PPLLEN bit is 0 (PPLL is operating) is prohibited.

PPLIDIV[1:0] Bits (PPLL Input Pulse Frequency Division Ratio Select)

The PPLIDIV[1:0] bits select the input pulse frequency divisor of the clock source for the PPLL.

Set the PPLIDIV[1:0] bits to bring the value within the range of input frequency for the PPLL (8 MHz to 24 MHz).

PPLSTC[5:0] Bits (PPLL Frequency Multiplier Setting)

The PPLSTC[5:0] bits set the frequency multiplier for the PPLL.

Set the PPLSTC[5:0] bits to bring the value within the range of output frequency for the PPLL (120 MHz to 240 MHz).

9.2.24 PPLL Control Register 2 (PPLLCR2)

Address(es): 0008 004Ah

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	PPLLEN
0	0	0	0	0	0	0	1

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	PPLLEN	PPLL Stop Control	0: PPLL is operating. 1: PPLL is stopped.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

PPLLEN Bit (PPLL Stop Control)

The PPLLEN bit controls operating and stopping of the PPLL.

The PPLL clock source is selectable as either the main clock oscillator or HOCO. The clock source is common with that of the PLL. Set the clock source in the PLLCR.PLLSRCSEL bit.

Selecting the main clock oscillator as the PPLL clock source in the PLLCR.PLLSRCSEL bit requires setting of the main clock oscillator wait control register (MOSCWTCR).

Start by setting the PPLLEN bit so that the PPLL operates, confirm that the OSCOVFSR.PPLOVF flag is 1, and then start using the PPLL clock.

The PPLL requires a certain amount of time for its oscillation to become stable after it is started. It also requires a certain amount of time for oscillation to stop after it has been stopped. For these reasons, the following restrictions apply to using the PPLLEN bit to start or stop the PPLL.

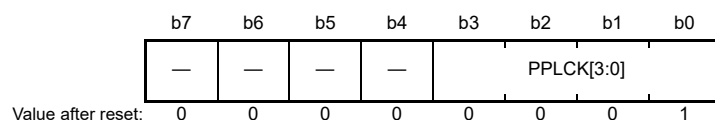
Notes on selecting the main clock oscillator as the PPLL clock source are listed below.

- Setting the PPLLEN bit to start the PPLL is possible regardless of the value of the OSCOVFSR.PPLOVF flag. However, regarding the time for processing to actually stop oscillation (that is, the time taken for the setting of the PPLOVF flag to become 0 after the setting to stop operation), compared with the setting to start operation when oscillation is stopped, writing to the PPLLCR2 register takes time.
- Stopping the PPLL by setting the PPLLEN bit can be done regardless of the value of the OSCOVFSR.PPLOVF flag. Writing to the PPLLCR2 register to make oscillation stop during the waiting time for it to become stable (that is, the time taken for the PPLOVF flag to be set to 1 after the setting to start operation) takes longer than the time for the processing to make the PPLL actually stop after making the setting during stable oscillation.
- To place the chip on software standby or deep software standby after making the setting to start the PPLL, start by confirming that the value of the OSCOVFSR.PPLOVF flag is 1, and only then execute the WAIT instruction.
- To place the chip on software standby or deep software standby after making the setting for the PPLL to stop, start by confirming that the value of the OSCOVFSR.PPLOVF flag is 0, and only then execute the WAIT instruction.

Writing 0 to the PPLLEN bit (making the PPLL operate) while low-speed mode 1 or 2 is selected by the operating power control mode select bits in the operating power control register (OPCCR.OPCM[2:0]) is prohibited.

9.2.25 PPLL Control Register 3 (PPLLCR3)

Address(es): 0008 004Bh



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	PPLCK[3:0]	PPLL Frequency-Divided Clock (PPLLDIVCLK) Select	b3 b0 0 0 0 1: ×1/2 0 0 1 0: ×1/3 0 0 1 1: ×1/4 0 1 0 0: ×1/5 Settings other than above are prohibited.*1	R/W
b7 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. When PPLLDIVCLK is selected as the UCLK for use with the USB, do not set values other than those listed above.
 When PPLLDIVCLK is selected as the UCLK and the USB is not to be used, these bits are read as 0001. When writing, write 0001b to them.

PPLCK[3:0] Bits (PPLL Frequency-Divided Clock (PPLLDIVCLK) Select)

The PPLCK[3:0] bits select the frequency of the PPLL frequency-divided clock (PPLLDIVCLK). The PPLLDIVCLK clock can be selected as the USB clock (UCLK). To make these selections, refer to [section 9.2.22, Specific-Use Clock Control Register \(PACKCR\)](#).

When division by 3 is selected, the clock duty cycle becomes 2 to 1. When division by 5 is selected, the clock duty cycle becomes 3 to 2.

Changing the value of the PPLLCR3 register while PPLLCR2.PPLLEN is 1 (the PPLL is stopped) is prohibited.

Start by writing 0 to PPLCR2.PPLLEN (making the PPLL operate), confirm that OSCOVFSR.PPLOVF is 1 (oscillation is stable), and then change the value in the PPLLCR3 register.

New settings for the bits also must only be made while supply of the clock signal to the USB module is stopped (by setting the MSTPCRB.MSTPB19 bit to 1).

9.3 Main Clock Oscillator

There are two ways of supplying the clock signal to the main clock oscillator: connecting an oscillator or the input of an external clock signal.

9.3.1 Connecting a Crystal Resonator

Figure 9.4 shows an example of connecting a crystal.

Connect capacitors referring to the capacitive load of the crystal to be used. In addition, a damping resistor R_d should be added, if necessary. The values of capacitors and resistor vary depending on the resonator and the oscillator driving ability. Use values recommended by the resonator manufacturer. If use of an external feedback resistor (R_f) is directed by the resonator manufacturer, insert an R_f between EXTAL and XTAL by following the instruction.

When connecting a resonator to supply the clock, the frequency of the crystal must be in the frequency range of the resonator for the main clock oscillator described in Table 9.1.

When a resonator is connected, setting the MOFCR.MODRV2[1:0] bits (Main Clock Oscillator Driving Ability 2 Switching) is required.

The frequency ranges that are specified in the bit description of the MODRV2[1:0] bits are the reference values of the crystal with capacitive load of $C_L = 8$ pF. The setting value may not fit within the frequency range depending on a crystal. Use values recommended by the resonator manufacturer.

In case of a ceramic resonator, it may be better to select lower frequency range than the frequency of the resonator. (For example, specify 10b instead of 01b when a ceramic resonator with the frequency range of 16.1 to 20 MHz is used). Use values recommended by the resonator manufacturer.

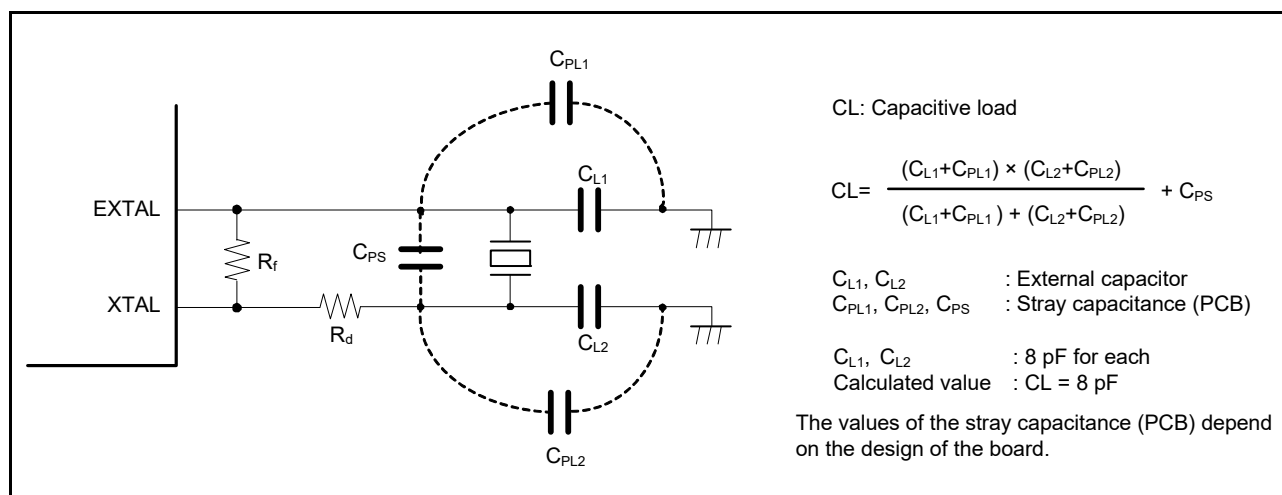


Figure 9.4 Example of Crystal Connection

Table 9.4 Damping Resistance (Reference Values)

Frequency (MHz)	8	12	16	20	24
R_d (Ω)	0	0	0	0	0

Figure 9.5 shows an equivalent circuit of the crystal resonator. Use a crystal resonator that has the characteristics listed in Table 9.5.

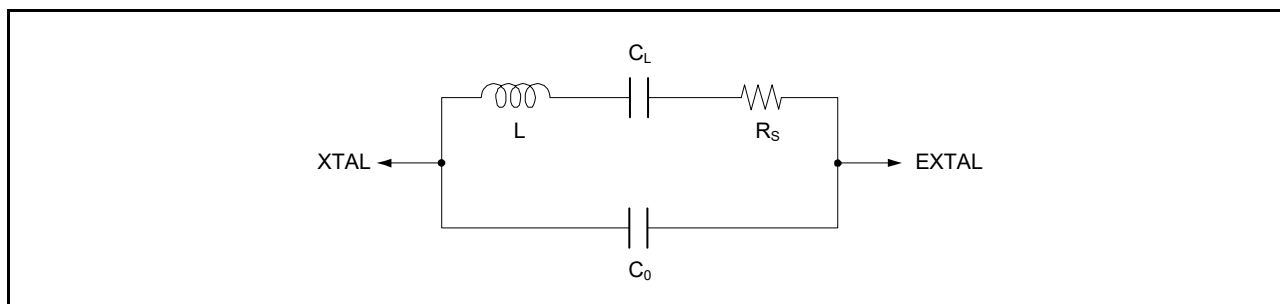


Figure 9.5 Equivalent Circuit of Crystal Resonator

Table 9.5 Crystal Characteristics (Reference Values)

Frequency (MHz)	8	12	16	20	24
R_s max (Ω)	300	100	80	50	50

9.3.2 External Clock Input

Figure 9.6 shows examples of connection of external clock input. Set the MOFCR.MOSEL bit to 1 and open the XTAL pin to operate the oscillator by inputting an external clock signal.

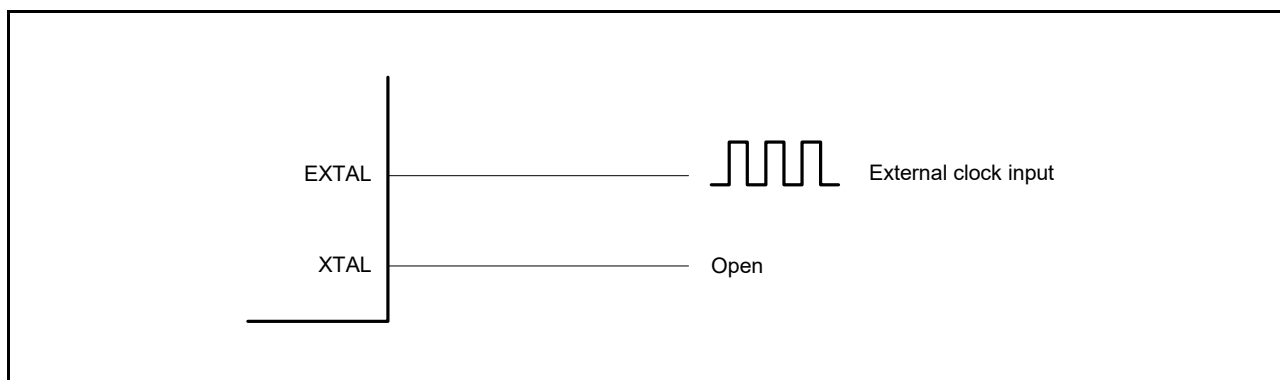


Figure 9.6 Example of Connection of External Clock

9.3.3 Notes on the External Clock Input

The frequency of the external clock input can only be changed while the main clock oscillator is stopped. Do not change the frequency of the external clock input while the setting of the main clock oscillator stop bit (MOSCCR.MOSTP) is 0 (making the main clock oscillator run) or that of the main clock oscillator forced oscillation bit (MOFCR.MOFXIN) is 1 (forcing the main clock oscillator to run).

9.4 Sub-Clock Oscillator

To supply a clock to the sub-clock oscillator, connect a crystal resonator.

9.4.1 Connecting 32.768-kHz Crystal Resonator

To supply a clock to the sub-clock oscillator, connect a 32.768-kHz crystal resonator, as shown in Figure 9.7.

A damping resistor R_d should be added, if necessary. Since the resistor values vary depending on the resonator and the oscillation driving ability, use values recommended by the resonator manufacturer. If use of an external feedback resistor (R_f) is directed by the resonator manufacturer, insert an R_f between XCIN and XCOU by following the instruction.

When connecting a resonator to supply the clock, the frequency of the resonator should be in the frequency range of the resonator for the sub-clock oscillator described in Table 9.1.

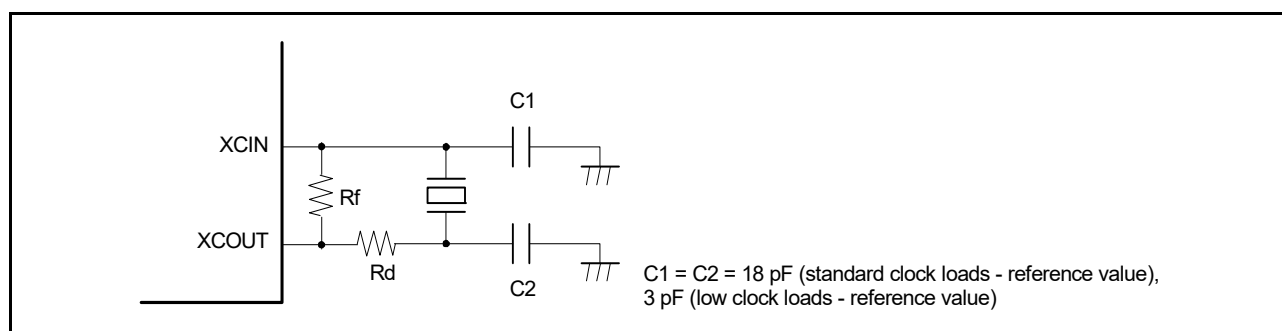


Figure 9.7 Connection Example of 32.768-kHz Crystal Resonator

Figure 9.8 shows an equivalent circuit for the 32.768-kHz crystal resonator. Use a crystal resonator that has the characteristics listed in Table 9.6.

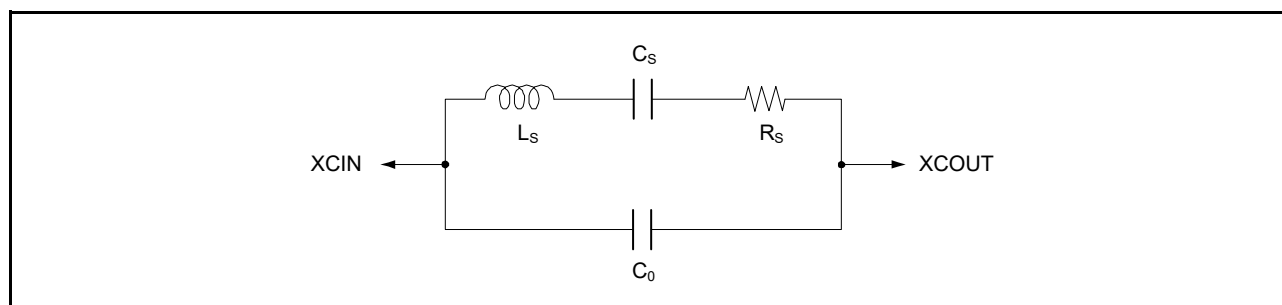


Figure 9.8 Equivalent Circuit for Crystal Resonator

Table 9.6 Crystal Resonator Characteristics (Reference Values)

Frequency (kHz)	32.768 (Low Clock Loads)	32.768 (Standard Clock Loads)
R_S max (k Ω)	60	60

9.4.2 Handling of Pins when Sub-Clock is Not Used

If the sub-clock is not in use, connect the XCIN pin to VSS via a resistor (to pull VSS down) and leave the XCOU pin open-circuit as shown in Figure 9.9.

In addition, if an oscillator is not connected, set the sub-clock oscillator stop bit (SOSCCR.SOSTP) to 1 (stopping the oscillator) and the sub-clock oscillator control bit in RTC control register 3 (RCR3.RTCEN) to 0 (stopping the sub-clock oscillator). The value of some RTC registers related to the sub-clock will be undefined after a cold start. Accordingly, be sure to set these bits after a cold start.

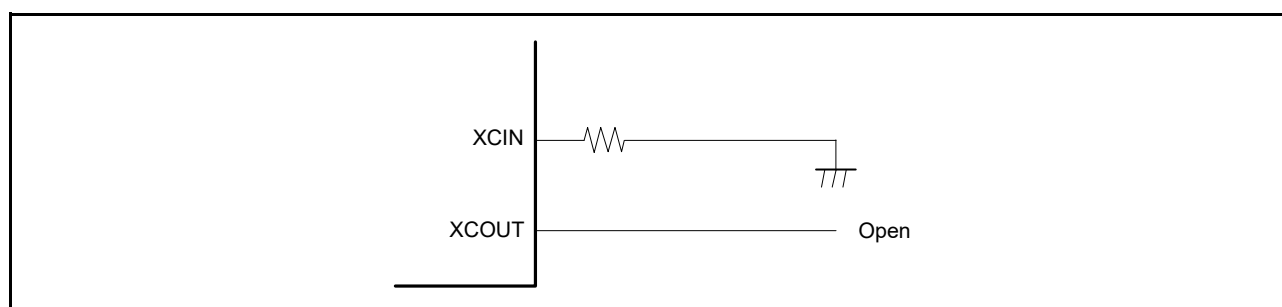


Figure 9.9 Pin Handling when Sub-Clock is Not Used

9.5 Oscillation Stop Detection Function

9.5.1 Oscillation Stop Detection and Operation after Detection

The oscillation stop detection function is used to detect the main clock oscillator stop and to supply LOCO clock pulses from the low-speed on-chip oscillator as the system clock source instead of the main clock or PLL clock. When the HOCO is selected as the clock source for the PLL and the PLL clock is selected as the system clock, the system clock is not switched to the LOCO even if stopping of the main clock is detected.

An oscillation stop detection interrupt request can be generated when an oscillation stop is detected. In addition, the MTU and GPTW output can be forcedly driven to the high-impedance on the detection. For details, refer to [section 24, Multi-Function Timer Pulse Unit 3 \(MTU3a\)](#), [section 25, Port Output Enable 3 \(POE3a\)](#), [section 26, General PWM Timer \(GPTW\)](#), and [section 27, GPTW Port Output Enable \(POEG\)](#).

In the MCU, the input clock remaining at a given level over a certain period due, for example, to a malfunction of the main clock oscillator, is the criterion to detect stopping of the main clock. For details of the detection period, refer to [Table 63.59, Oscillation Stop Detection Circuit Characteristics](#).

When an oscillation stop is detected, the main clock or PLL clock selected by the clock source select bits (SCKCR3.CKSEL[2:0]) is switched to the LOCO clock by the corresponding selectors in the former stage.

Therefore, on detection of oscillation stopping while the PLL clock or system clock is selected as the source of the main clock, the system clock source is switched to the LOCO clock without a change of CKSEL[2:0].

Even when stopping of oscillation is detected while the PPLL frequency-divided clock is selected by the PACKCR.UPLLSEL bit, the setting of the PACKCR.UPLLSEL bit does not change and the source of the UCLK remains the PPLL clock. Note that the frequency becomes a special frequency (the frequency of free-running oscillation).

Switching between the main clock and LOCO clock or between the PLL clock and LOCO clock is controlled by the oscillation stop detection flag (OSTDSR.OSTDF). The clock source is switched to the LOCO clock when the OSTDF flag is 1, and is switched to the main clock or PLL clock again when the OSTDF flag is set to 0. At this time, if the main clock or PLL clock is selected with the CKSEL[2:0] bits, the OSTDF flag cannot be set to 0. To switch the clock source to the main clock or PLL clock again after the oscillation stop detection, set the CKSEL[2:0] bits to a clock source other than the main clock or PLL clock and clear the OSTDF flag to 0. After that, check that the OSTDF flag is not 1, and then set the CKSEL[2:0] bits to the main clock or PLL clock after the specified oscillation settling time has elapsed.

After a reset is released, the main clock oscillator is stopped and the oscillation stop detection function is disabled. To enable the oscillation stop detection function, activate the main clock oscillator and write 1 to the oscillation stop detection function enable bit (OSTDCR.OSTDE) after confirming that the OSCOVFSR.MOOVF flag or OSCOVFSR.PLOVF flag have been set to 1.

The oscillation stop detection function is provided against the main clock stop by an external cause. Therefore, the oscillation stop detection function should be disabled before the main clock oscillator is stopped by the software or a transition is made to software standby mode or deep software standby mode.

The clocks that are switched to the LOCO clock by the oscillation stop detection are: the main clock, PLL clock, CAC main clock (CACMCLK), and CAN clock (CANMCLK), which are provided as the system clock sources. The main clock as the RTC main clock source (RTCMCLK) is not switched to the LOCO clock.

Note that the frequencies of the derived clock signals after switching to the LOCO depends on the settings of the system clock control registers (SCKCR, SCKCR2, or SCKCR3).

Figure 9.10 shows an example of a flowchart for initialization of the oscillation stop detection function.

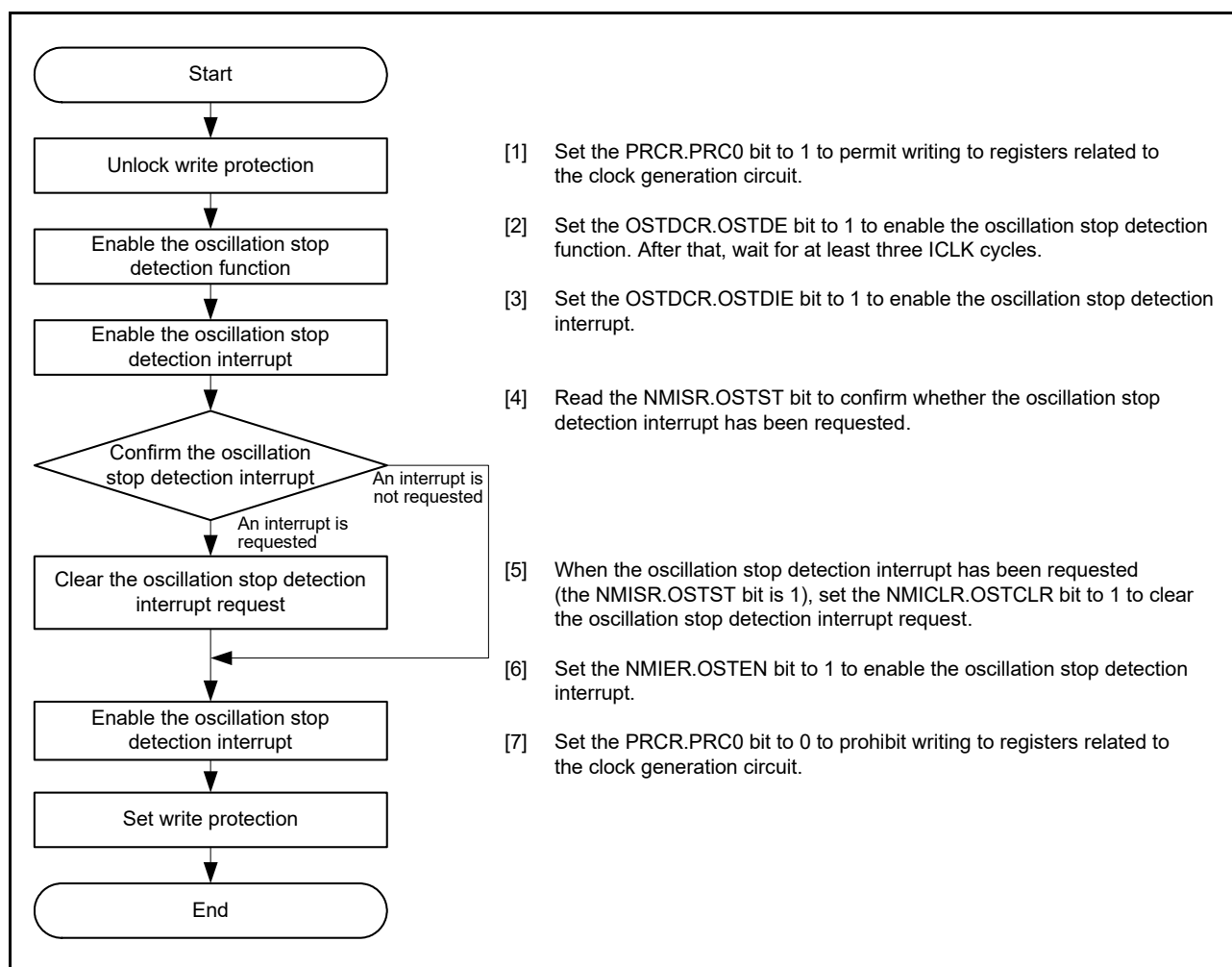


Figure 9.10 Flowchart Example for Initialization of Oscillation Stop Detection Function

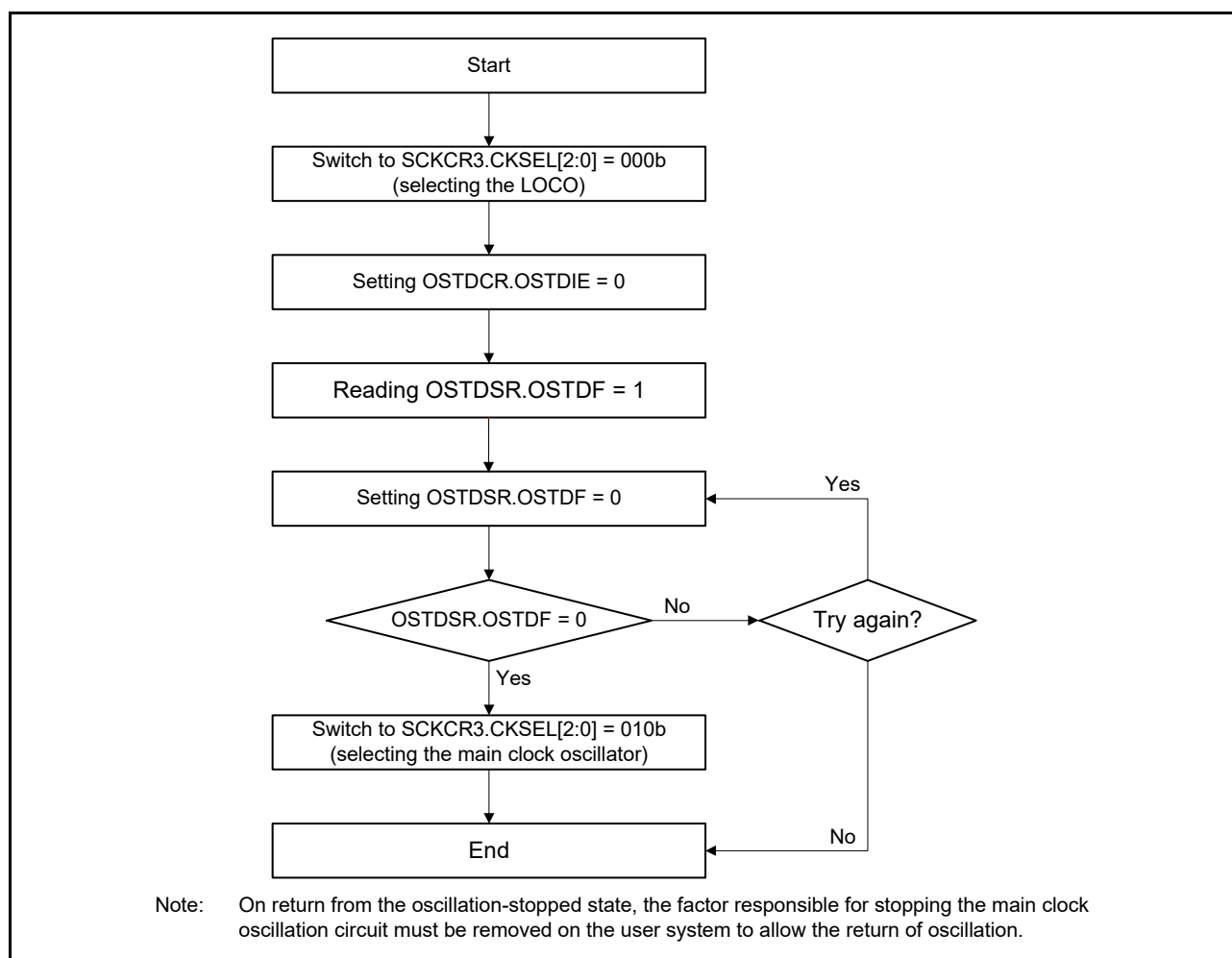


Figure 9.11 Flowchart Example for Recovery from Detection of Oscillator Stop

9.5.2 Oscillation Stop Detection Interrupts

An oscillation-stop detection interrupt (OSTDI) will be generated if the oscillation-stop detection flag (OSTDSR.OSTDF) becomes 1 while the oscillation-stop detection interrupt enable bit in the oscillation stop detection control register (OSTDCR.OSTDIE) is 1 (enabling interrupt generation on oscillation stop detection). At this time, the port output enable 3 (POE) and the GPTW port output enable (POEG) modules are notified that the main clock oscillator has stopped. On accepting the notification of the oscillation stop, the POE sets the OSTST high-impedance flag in input level control/status register 6 (ICSR6.OSTSTF) to 1. After the oscillation stop is detected, wait for at least 10 cycles of PCLKB before writing to this ICSR6.OSTSTF flag. When the OSTDSR.OSTDF flag requires clearing, do so after clearing the oscillation-stop detection interrupt enable bit in the oscillation stop detection control register (OSTDCR.OSTDIE). Wait for at least two cycles of PCLKB clock before again setting the OSTDCR.OSTDIE bit to 1. According to the number of cycles for access to read out a given I/O register, waiting time longer than two cycles of PCLKB may have to be secured.

In response to the notification of the main clock oscillator having stopped, the POEG also sets the oscillation stop detection flags (POEGn.OSTPF (n = A to D)) in the POEG group n setting registers to 1.

If the oscillation stop detection interrupt is to be used as a non-maskable interrupt, since non-maskable interrupts are disabled in the initial state after a reset release, set the corresponding bit in the NMIER register to 1 by software to enable non-maskable interrupts. If it is to be used as a maskable interrupt, do not change the value of the NMIER register from the value after a reset. For details, refer to section 15, Interrupt Controller (ICUD).

9.6 PLL Circuit

The PLL and PPLL circuits are capable of multiplying the frequency of the clock signal from the oscillator.

9.7 Internal Clock

Clock sources of internal clock signals are the main clock, sub-clock, HOCO clock, LOCO clock, PLL clock, PPLL clock, dedicated clock for the IWDT, and the JTAG external clock. The internal clocks listed in Table 9.7 are produced from these sources.

Frequencies of the internal clocks are set by the combination of the divisors selected by the FCK[3:0], ICK[3:0], BCK[3:0], PCKA[3:0], PCKB[3:0], PCKC[3:0], and PCKD[3:0] bits in SCKCR, the BCLKDIV bit in BCKCR, the UCK[3:0] bits in SCKCR2, PPLCK[3:0] bits in PPLLCR3, CKODIV[2:0] bits in CKOCR, the clock source selected by the CKSEL[2:0] bits in SCKCR3, the PLLSRCSEL bit in PLLCR, the EPLLSEL, UPLLSEL and OUTCKSEL bit in PACKCR, the bits that select the frequency of the PLL circuit (STC[5:0] and PLIDIV[1:0] in PLLCR), the HCFRQ[1:0] bits in HOCOCR2, and the bits that select the frequency of the PPLL circuit (PPLSTC[5:0] and PPLIDIV[1:0] in PPLLCR). If the value of any of these bits is changed, subsequent operation will be at the frequency determined by the new value.

Table 9.7 Internal Clocks and Supply Destination Modules (1/2)

	Type of Internal Clock	Clock Name	Supply Destination Module
1	System clock	ICLK	CPU, code flash memory, RAM, ICU, BSC, DMAC, DTC, EXDMAC, MPU
2	Peripheral module clocks	PCLKA	MTU, SCI7 to SCI11, RSPI, ETHERC, EDMAC, EPTPC, GLCDC, DRW2D, PMGI, GPTW
		PCLKB	TPU, PPG, TMR, CMT, CMTW, RTC, WDT, IWDT, SCI12, SCI0 to SCI6, RIIC, CAN, USB, SDHI, MMCIF, QSPI, PDC, S12AD, R12DA, TEMPS, CRC, DOC, CAC, TSIP, standby RAM, I/O, MPC, ICU, POE3, POEG, SSIE
		PCLKC	S12AD (unit 0)
		PCLKD	S12AD (unit 1)
3	Flash-IF clock	FCLK	Data flash memory, code flash memory
4	External bus clock	BCLK	BSC, I/O
5	SDRAM clock	SDCLK	I/O
6	USB clock	UCLK	USB
7	Ethernet-PHY external clock	CLKOUT25M	I/O
8	CLKOUT clock	CLKOUT	I/O
9	CAN clock	CANMCLK	CAN
10	CAC clocks	CACMCLK (Main clock)	CAC
		CACCLK (Sub clock)	
		CACHCLK (HOCO clock)	
		CACLCLK (LOCO clock)	
		CACILCLK (IWDT-dedicated clock)	
		CACUCLK (CACUSB clock)	
		CACCLKOUT25M (CACCLKOUT25M clock)	

Table 9.7 Internal Clocks and Supply Destination Modules (2/2)

	Type of Internal Clock	Clock Name	Supply Destination Module
11	RTC clocks	RTCMCLK (Main clock)	RTC
		RTCSCLK (Sub clock)	
12	IWDT-dedicated clock	IWDTCLK	IWDT, CAC
13	JTAG clock	JTAGTCK	Boundary scan

9.7.1 System Clock

The system clock (ICLK) is used as the operating clock of the CPU, DMAC, DTC, code flash memory, and RAM. The ICLK frequency is specified by the ICK[3:0] bits in SCKCR, the CKSEL[2:0] bits in SCKCR3, the PLLSRCSEL, STC[5:0], and PLIDIV[1:0] bits in PLLCR, and the HCFRQ[1:0] bits in HOCOCR2.

9.7.2 Peripheral Module Clock

The peripheral module clocks (PCLKA, PCLKB, PCLKC, and PCLKD) are the operating clocks for use by peripheral modules.

The frequency of the given clock is specified by the PCKA[3:0], PCKB[3:0], PCKC[3:0], and PCKD[3:0] bits in SCKCR, the CKSEL[2:0] bits in SCKCR3, the PLLSRCSEL, STC[5:0], and PLIDIV[1:0] bits in PLLCR, and the HCFRQ[1:0] bits in HOCOCR2. The peripheral module clocks can be set to frequencies above that of the system clock.

9.7.3 Flash-IF Clock

The flash-interface clock (FCLK) is used as the operating clock for the flash-memory interfaces. That is, FCLK is used for programming and erasure of the code flash memory and data flash memory, and reading from the data flash memory. The FCLK frequency is specified by the FCK[3:0] bits in SCKCR, the CKSEL[2:0] bits in SCKCR3, and the PLLSRCSEL, STC[5:0], and PLIDIV[1:0] bits in PLLCR, and the HCFRQ[1:0] bits in HOCOCR2.

9.7.4 External Bus Clock

The external bus clock (BCLK) is an operating clock for the external bus controller and EXDMAC. It is also output externally from the BCLK pin to the external bus. When the external bus is enabled, P53 that is function-multiplexed with the BCLK pin cannot be used as an I/O port.

BCLK can be output from the BCLK pin by setting the SCKCR.PSTOP1 bit to 0 and setting the external bus enable bit in the system control register 0 (SYSCR0.EXBE) to 1. Make sure that modification of the SYSCR0.EXBE bit to 1 must always be performed while the PSTOP1 bit in SCKCR is 1.

When the BCKCR.BCLKDIV bit is set to 1, the BCLK clock divided by 2 is output from the BCLK pin.

The BCLK frequency is specified by the BCK[3:0] bits in SCKCR, the CKSEL[2:0] bits in SCKCR3, and the PLLSRCSEL, STC[5:0], and PLIDIV[1:0] bits in PLLCR, and the HCFRQ[1:0] bits in HOCOCR2.

A frequency higher than the system clock (ICLK) should not be set for the BCLK.

9.7.5 SDRAM Clock

The SDRAM clock (SDCLK) is an operating clock for the external bus controller. It is output externally from the SDCLK pin for the SDRAM that is connected to the external bus.

When the SCKCR.PSTOP0 bit is set to 0 and the SDCLK enable bit (SDCLKE) in the external bus control register 1 (PFBCR1) is set to 1 (enabling SDCLK output), it selects output of SDCLK on the SDCLK pin. When changing the value of the PFBCR1.SDCLKE bit, make sure that the value of the SCKCR.PSTOP0 bit is 1.

The SDCLK clock frequency is specified by the SCKCR.BCK[3:0], SCKCR3.CKSEL[2:0], PLLCR.PLLSRCSEL, STC[5:0], and PLIDIV[1:0], and HOCOCR2.HCFRQ[1:0] bits.

A frequency higher than the system clock (ICLK) should not be set for the SDCLK

9.7.6 USB Clock

The USB clock (UCLK) is used as the operating clock for USB.

The settings of the following bits determine the frequency of the UCLK:

SCKCR2.UCK[3:0], SCKCR3.CKSEL[2:0], PLLCR.PLLSRCSEL, STC[5:0], PLIDIV[1:0], PPLLCR3.PPLCK[3:0], PACKCR.UPLLSEL, PPLLCR.PPLSTC[5:0], and PPLIDIV[1:0]. The UCLK frequency must be set to 48 MHz.

9.7.7 Ethernet-PHY External Clock

The Ethernet-PHY external clock (CLKOUT25M) is the external output clock for the Ethernet-PHY chip. CLKOUT25M can be externally output from the CLKOUT25M pin.

The settings of the following bits determine the frequency of the CLKOUT25M:

PLLCR.PLLSRCSEL, PACKCR.OUTCKSEL, PLLCR.PLIDIV[1:0], STC[5:0], PPLLCR.PPLIDIV[1:0], PPLSTC[5:0], and HOCOCR2.HCFRQ[1:0].

A clock signal produced by frequency-dividing the signal from the PLL or PPLL by 8 is generated. The settings are required to result in a frequency of 25 MHz for the CLKOUT25M.

9.7.8 CLKOUT Clock

The CLKOUT clock (CLKOUT) signal can be externally output from the CLKOUT pin.

The settings of the CKOCR.CKSEL[2:0] and CKODIV[2:0] bits determine the frequency of the CLKOUT signal.

9.7.9 CAN Clock

The CAN clock (CANMCLK) is an operating clock for the CAN.

CANMCLK is generated by the main clock oscillator.

9.7.10 CAC Clock (CACCLK)

The CAC clock (CACCLK) is an operating clock for the CAC.

The CACCLK is selectable from among

CACMCLK generated by the main clock oscillator; CACSCLK generated by the sub-clock oscillator; CACHCLK generated by the high-speed on-chip oscillator; CACLCLK generated by the low-speed on-chip oscillator; CACILCLK generated by the IWDT on-chip oscillator; PCLKB that is provided to peripheral modules; CACUCLK that is provided as the USB clock; and CACCLKOUT25M that is the Ethernet-PHY external clock.

9.7.11 RTC Clock

The RTC clock (RTCSCLK, RTCMCLK) is the operating clock for the RTC.

RTCSCLK is generated by the sub-clock oscillator, and RTCMCLK is generated by the main clock oscillator.

9.7.12 IWDT-Dedicated Clock

The IWDT-dedicated clock (IWDTCCLK) is the operating clock for the IWDT.

IWDTCCLK is internally generated by the IWDT-dedicated on-chip oscillator.

9.7.13 JTAG Clock

The JTAG clock (JTAGTCK) is the operating clock for the JTAG.

JTAGTCK is generated by the JTAG external clock (TCK).

9.8 Clock Source Switching

In this MCU, the clock signal from the LOCO, which oscillates during release from the reset state, is used to start the fetching of CPU instructions after the internal reset time (tRESWT) has elapsed. After that, set up the clock to which the CPU will be switched while it is still driven by the LOCO, and read the oscillation stabilization flag register to confirm that oscillation of the given clock signal is stable before switching to the selected clock source.

- (1) Example of procedure for settings to switch the system clock source from the LOCO to the PLL (clock source for the PLL: main clock) after release from an internal reset
 1. After release from the internal reset state, set the driving ability by writing to the MODRV2[1:0] bits in the MOFCR register.
 2. Set the oscillation settling time for the main clock oscillator by writing to the MSTs[7:0] bits in the MOSCWTCR register.
 3. Make the main clock oscillator run by setting the MOSTP bit in the MOSCCR register.
 4. When ICLK is to be set to run at a frequency higher than 120 MHz, set the MEMWAIT bit in the MEMWAIT register to 1.
 5. Set the frequency multiplication factor in the PLLCR register (the initial setting for the PLL clock source selects the main clock oscillator).
 6. Select PLL operation by writing to the PLL stop control bit in the PLLCR2 register.
 7. Confirm that the PLL clock has become stable by reading the PLOVF flag in the OSCOVFSR register.
 8. Change the clock signal from the LOCO clock to the PLL clock by writing to the CKSEL[2:0] bits in the SCKCR3 register.

- (2) Example of procedure for settings to switch the system clock source from the LOCO to the PLL (clock source for the PLL: HOCO) after release from an internal reset
 1. After release from the internal reset state, set the frequency by writing to the HCFRQ[1:0] bits in the HOCOCR2 register.
 2. Make the HOCO clock run by setting the HCSTP bit in the HOCOCR register (the initial value depends on the value of the OFS1.HOCOEN bit; if the OFS1.HOCOEN bit is 0, an explicit setting to make the HOCO clock run is not required. The oscillation settling time for the HOCO is 25 cycles of the LOCO).
 3. When ICLK is to be set to run at a frequency higher than 120 MHz, set the MEMWAIT bit in the MEMWAIT register to 1.
 4. Set the frequency multiplication factor and set the HOCO clock as the PLL clock source by writing to the PLLCR register.
 5. Select PLL operation by writing to the PLL stop control bit in the PLLCR2 register.
 6. Confirm that the PLL clock has become stable by reading the PLOVF flag in the OSCOVFSR register.
 7. Change the clock signal from the LOCO clock to the PLL clock by writing to the CKSEL[2:0] bits in the SCKCR3 register.

- (3) Example of procedure for settings to switch the system clock source from the LOCO to the main clock after release from an internal reset
 1. After release from the internal reset state, set the driving ability by writing to the MODRV2[1:0] bits in the MOFCR register.
 2. Set the oscillation settling time for the main clock oscillator by writing to the MSTs[7:0] bits in the MOSCWTCR register.
 3. Make the main clock oscillator run by setting the MOSTP bit in the MOSCCR register.
 4. Confirm that the main clock has become stable by reading the MOOVF flag in the OSCOVFSR register.
 5. Change the clock signal from the LOCO clock to the main clock by writing to the CKSEL[2:0] bits in the SCKCR3 register.

- (4) Example of procedure for settings to switch the system clock source from the LOCO to the HOCO after release from an internal reset
 1. After release from the internal reset state, set the frequency by writing to the HCFRQ[1:0] bits in the HOCOCCR2 register.
 2. Make the HOCO clock run by setting the HCSTP bit in the HOCOCCR register (the initial value depends on the value of the OFS1.HOCOEN bit; if the OFS1.HOCOEN bit is 0, an explicit setting to make the HOCO clock run is not required. The oscillation settling time for the HOCO is 25 cycles of the LOCO).
 3. Confirm that the HOCO clock has become stable by reading the HCOVF flag in the OSCOVFSR register.
 4. Change the clock signal from the LOCO clock to the HOCO clock by writing to the CKSEL[2:0] bits in the SCKCR3 register.

- (5) An example of settings to change the system clock from that from the LOCO to that from the PLL (clock source: main clock) and of the PPLL frequency-divided clock being selected as the USB clock after release from an internal reset is given below.
 1. After release from the internal reset state, set the driving ability by writing to the MODRV2[1:0] bits in the MOFCR register.
 2. Set the oscillation settling time for the main clock oscillator by writing to the MSTS[7:0] bits in the MOSCWTCR register.
 3. Make the main clock oscillator run by setting the MOSTP bit in the MOSCCR register.
 4. Set the source of the UCLK as the PPLLDIVCLK in the UPLLSEL bit of the PACKCR register.
 5. When ICLK is to be set to run at a frequency higher than 120 MHz, set the MEMWAIT bit in the MEMWAIT register to 1.
 6. Set the frequency multiplication factor in the PLLCR register (the initial setting for the PLL clock source selects the main clock oscillator).
 7. Select PLL operation by writing to the PLL stop control bit in the PLLCR2 register.
 8. Confirm that the PLL clock has become stable by reading the PLOVF flag in the OSCOVFSR register.
 9. Change the clock signal from the LOCO clock to the PLL clock by writing to the CKSEL[2:0] bits in the SCKCR3 register.
 10. Set the frequency multiplier in the PPLLCR register.
 11. Select PPLL operation by writing to the PPLL stop control bit in the PPLLCR2 register.
 12. Confirm that the PPLL clock has become stable by reading the PPLOVF flag in the OSCOVFSR register.
 13. Set the frequency divisor for the PPLLDIVCLK in the PPLLCR3 register.
 14. Change the setting of the MSTPB19 bit in the MSTPCRB register to release the USB module from the stopped state.

9.9 Operations Linked by the ELC

9.9.1 Event Signal Output to the ELC

The clock generation circuit is capable of operation linked with another module set in advance when its interrupt request signal is used as an even signal by the event link controller (ELC) on detection of stopping of the main clock oscillation. The clock generation circuit outputs the event signal regardless of the setting of the corresponding oscillation stop detection interrupt enable bit (OSTDCR.OSTDIE). For details, refer to [section 21, Event Link Controller \(ELC\)](#).

9.9.2 Clock Source Switching on Reception of the Event Signal from the ELC

The clock generation circuit is capable of switching the clock source to the low-speed on-chip oscillator in response to the event set in advance when the event specified in the ELSRn register of the ELC occurs.

While this function is in use, clock source switching on return from sleep mode cannot be used. For details, refer to [section 11.2.7, Sleep Mode Return Clock Source Switching Register \(RSTCKCR\)](#).

9.10 Usage Notes

9.10.1 Notes on Clock Generation Circuit

- (1) The frequencies of the system clock (ICLK), peripheral module clock (PCLKA to PCLKD), flash-IF clock (FCLK), external bus clock (BCLK), and SDRAM clock (SDCLK) supplied to each module change according to the settings of SCKCR. Each frequency should meet the following:
 Select each frequency that is within the operation guaranteed range of clock cycle time (tcyc) specified in AC characteristics of electrical characteristics.
 The frequencies must not exceed the ranges listed in Table 9.1.
 The peripheral modules operate on the PCLKB and PCLKA. Note therefore that the operating speed of modules such as the timer and SCI varies before and after the frequency is changed.
 Furthermore, PCLKC (unit 0) and PCLKD (unit 1) are available as clocks for the A/D converter.
 Do not set a higher frequency than PCLKB, which serves as the operating clock of the A/D converter.
- (2) The following relation is required between the frequencies of the system clock (ICLK) and external bus clock (BCLK).

$$\text{ICLK} : \text{BCLK} = N : 1 \text{ (N: integer) and } \text{ICLK} \geq \text{BCLK}.$$
 Also, the following relation is required between the frequencies of the peripheral module clocks.

$$\text{PCLKA} \geq \text{PCLKB}; \text{PCLKB} \geq \text{PCLKC}; \text{PCLKB} \geq \text{PCLKD}$$
- (3) Do not change the clock frequency during external bus access. Furthermore, when access via the external bus is to start after a change to the clock frequency, only start access via the bus after confirming that the change to the frequencies has been completed.
- (4) If the clock frequency is to be changed by modifying the value of SCKCR, SCKCR2, SCKCR3, or BCKCR register, wait for writing of the value to the register to be complete and the new frequency to be stable before starting subsequent processing. For the procedure to confirm the completion of writing to I/O registers, refer to (2) Notes on Writing to I/O Registers, in section 5, I/O Registers.

9.10.2 Note on Rewriting the SCKCR3 Register

When the SCKCR3.CKSEL[2:0] bits have been rewritten, the clock output is temporarily stopped to prevent the switch of the clock source from generating a clock pulse of short duration (glitch). The interrupt controller and event link controller may not detect the following signals that were input during this period.

- (1) An external pin interrupt or NMI pin interrupt with a pulse width shorter than 4 cycles of the PCLKB after the switch while the PCLKB frequency is the 1/1 of the clock source (the SCKCR.PCKB[3:0] bits are 0000b).
- (2) An external pin interrupt or NMI pin interrupt with a pulse width shorter than 2.5 cycles of the PCLKB after the switch while the PCLKB frequency is the 1/2 of the clock source (the SCKCR.PCKB[3:0] bits are 0001b).
- (3) An RTC periodic interrupt or RTC periodic event output when the SCKCR3.CKSEL[2:0] bits are changed to "011b" (sub-clock) while the count source of the RTC is set to the sub-clock (RCR4.RCKSEL bit is 0).
- (4) An RTC periodic interrupt or RTC periodic event output when the SCKCR3.CKSEL[2:0] bits are changed to the value other than "100b" (PLL circuit) while the count source of the RTC is set to the main clock (RCR4.RCKSEL bit is 1).

When the external pin interrupt or NMI pin interrupt is in use, input the signal with enough pulse width to exceed the time condition described in (1) and (2). When the RTC periodic interrupt or RTC periodic event output is in use, switch the clock source after a periodic interrupt is generated and before the next periodic interrupt is generated.

9.10.3 Notes on Resonator

Since various resonator characteristics relate closely to the user's board design, adequate evaluation is required on the user side before use, referencing the resonator connection example shown in this section. The circuit constants for the resonator depend on the resonator to be used and the stray capacitance of the mounting circuit. Therefore, the circuit constants should be determined in full consultation with the resonator manufacturer. The voltage to be applied between the resonator pins must be within the absolute maximum rating.

9.10.4 Notes on Board Design

When using a crystal resonator, place the resonator and its load capacitors as close to the XTAL and EXTAL pins as possible. Other signal lines should be routed away from the oscillation circuit as shown in Figure 9.12 to prevent electromagnetic induction from interfering with correct oscillation.

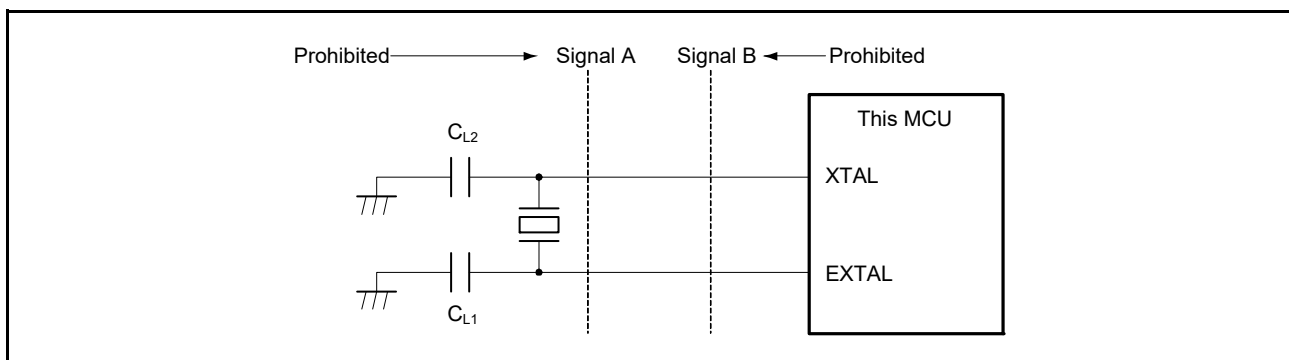


Figure 9.12 Notes on Board Design for Oscillation Circuit (Applies to the Sub-Clock Oscillator, in Case of the Main Clock Oscillator)

9.10.5 Notes on Resonator Connect Pin

When the main clock is not used, the EXTAL and XTAL pins can be used as general ports P36 and P37. When they are used as the general ports, the main clock should be stopped (MOSCCR.MOSTP should be set to 1 and MOFCR.MOFOXIN should be set to 0). However, with the system using the main clock, the EXTAL (P36) and XTAL (P37) pins should not be used as output ports.

For the values of registers related to port settings, refer to Table 23.37, Register Settings.

Furthermore, since the main clock becomes essential if the function indicated below is in use, design the board so that both pins can be used for the main clock signal.

- Programming of flash memory in boot mode (USB interface)*1

Note 1. For the conditions on the oscillator in the various modes, refer to section 62.20.7, Notes on Program Execution in Boot Mode (USB Interface).

9.10.6 Notes on Sub-Clock Oscillator

The sub-clock can be used as the system clock, as the source to drive counting by the realtime clock, or as both.

Accordingly, take note of the following limitations and points for caution regarding the settings, including when the sub-clock is not in use.

- To select the sub-clock as the system clock, set the SOSCCR.SOSTP bit; to set the sub-clock as the source to drive counting by the realtime clock, use the SOSCCR.SOSTP and RCR3.RTCEN bits. Furthermore, when the sub-clock is to be used as the clock source for counting by the realtime clock, the SOSCWTCR.SSTS[7:0] bits must be set to 00000000b after the oscillation stabilization waiting time has elapsed once the sub-clock has started oscillating.
- Make initial settings according to the example flowchart shown in Figure 9.13 when the sub-clock is to be used as the system clock as well as the source to drive counting by the realtime clock or when the sub-clock is only to be used as the source to drive counting by the realtime clock. And then, follow the clock setting procedure described in section 33.3.2, Clock and Count Mode Setting Procedure.

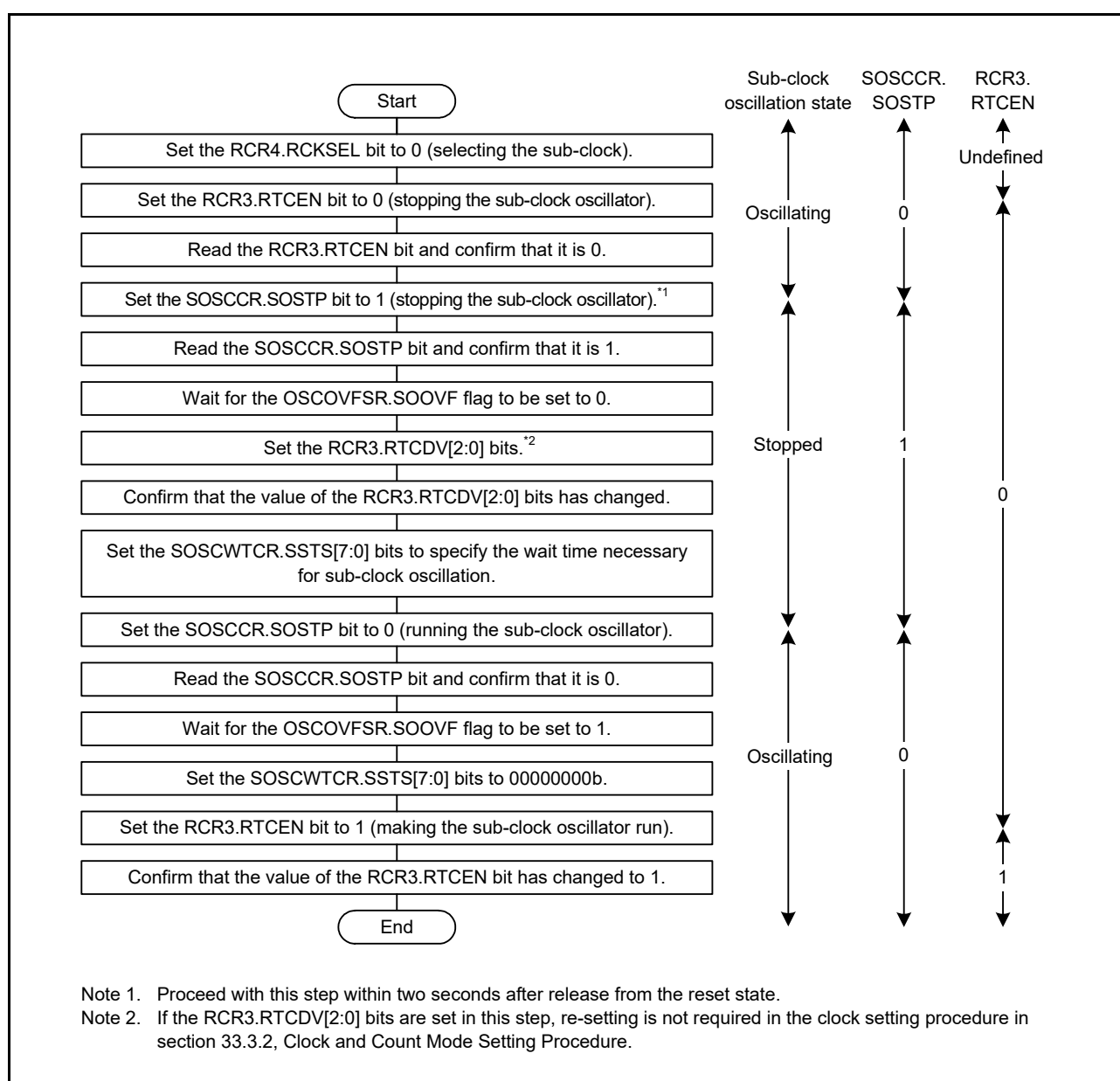


Figure 9.13 Example of Initialization Flowchart When Sub-Clock is Used as the Source to Drive Counting by the Realtime Clock

- Make initial settings according to the example flowchart shown in Figure 9.14 when the sub-clock is only to be used as the system clock.

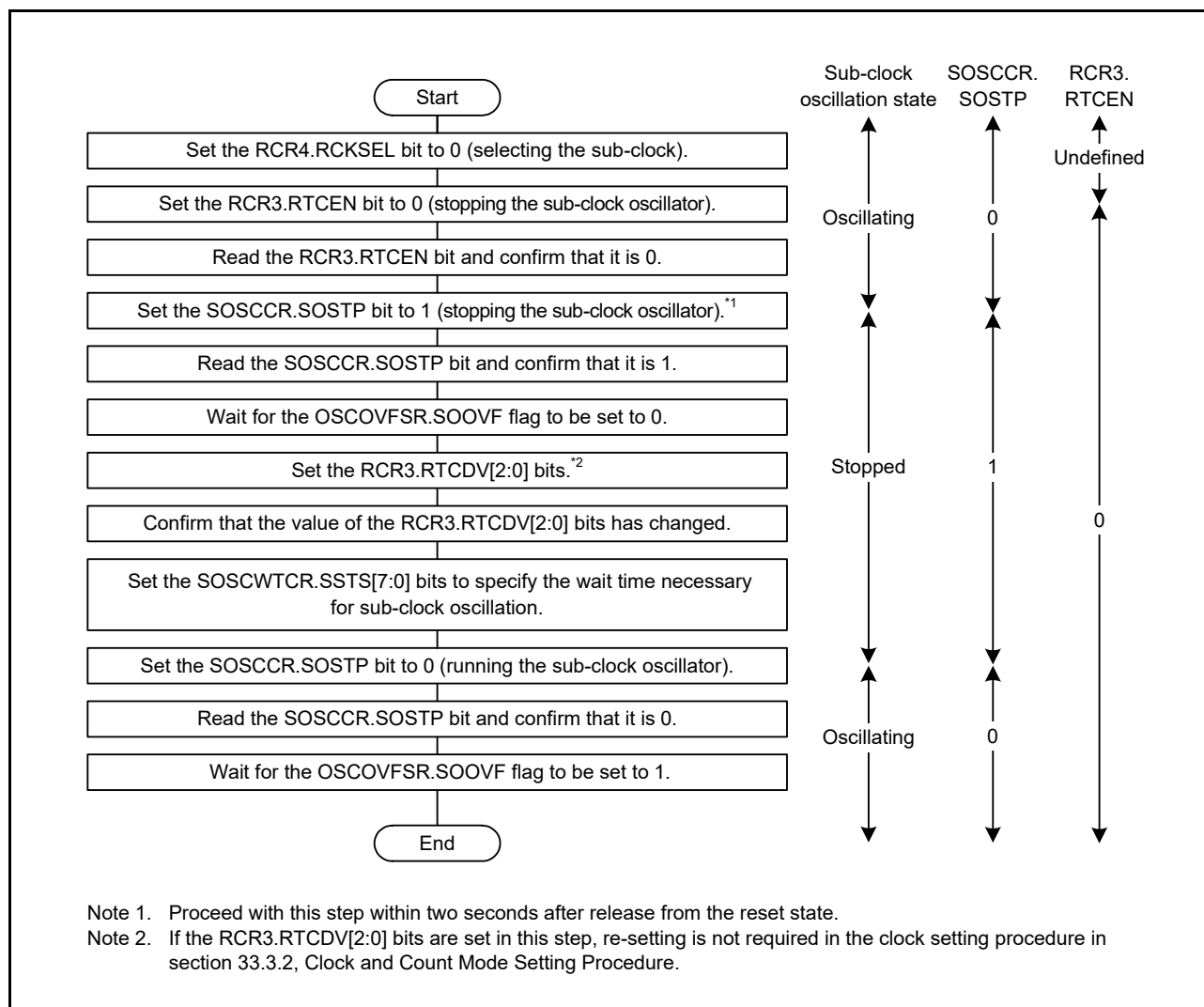


Figure 9.14 Example of Initialization Flowchart When Sub-Clock is Used Only as the System Clock

- Make initial settings according to the example flowchart shown in Figure 9.15 when the sub-clock is not used.

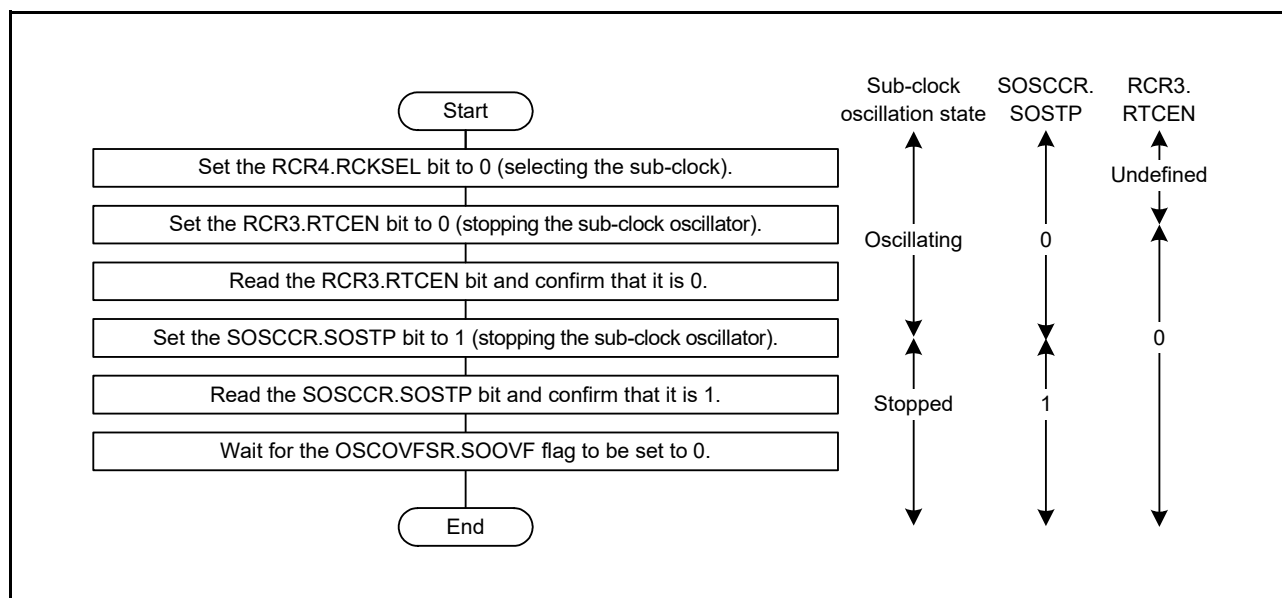


Figure 9.15 Example of Initialization Flowchart When Sub-Clock is Not Used

- When the sub-clock is used as the system clock, even if the RCR3.RTCEN bit is set to 1 and the sub-clock is already oscillating, wait for the OSCOVFSR.SOOVF flag to be set to 1 after the SOSCCR.SOSTP bit changes from 1 (stopped) to 0 (operating) before starting to use the sub-clock as the system clock.
- The state of the sub-clock control circuit is undefined after a cold start so the sub-clock must be initialized whether or not it is to be used. Setting both the SOSCCR.SOSTP bit and the RCR3.RTCEN bit to the “stopped” setting will initialize the control circuit. For initialization of the RCR3.RTCEN bit, refer to [section 33, Realtime Clock \(RTCd\)](#).
- For the sub-clock oscillator to operate, the RCR3.RTCDV[2:0] bits must also be set. This setting must be made while the sub-clock oscillator is stopped. Writing to these bits during operation is prohibited.
- If the RCR3.RTCEN bit is modified after the SOSCCR.SOSTP bit is modified, or if the SOSCCR.SOSTP bit is modified after the RCR3.RTCEN bit is modified, make sure that first modified bit has been rewritten before the subsequent bit is modified.

9.10.7 Notes on Using a Low CL Crystal Unit

When the RCR3.RTCDV[2:0] bits are 001b (drive capacity for low CL), the oscillator is susceptible to noise. Especially when the signal level of any pin near the XCIN or XCOUT pin is changed, the oscillation accuracy of the sub-clock oscillator may be affected. The accuracy is affected differently depending on the board traces and how the signal level of any pin near the XCIN or XCOUT pin is changed. When designing a board using a low CL crystal unit, refer to the application note “Design Guide for Low CL Sub-clock Circuits” (R01AN1187EJ) to reduce the influence from noise. The following are examples that may significantly affect oscillation accuracy:

(1) When connecting an on-chip debugging emulator to the FINED pin

Since the FINED pin (FINE interface pin) is near the XCIN and XCOUT pins, the oscillation accuracy of the sub-clock oscillator is affected when using the FINED pin in debugging. When using the FINED pin in debugging, set the RCR3.RTCDV[2:0] bits to 001b (drive capacity for low CL) and debug at the room temperature.

(2) When supplying an external clock to the main clock oscillator

When inputting an external clock to the EXTAL pin, the oscillation accuracy of the sub-clock oscillator may be affected. When inputting an inverted external clock to the XTAL pin, the oscillation accuracy will be affected more significantly.

9.10.8 Notes on Changing the ICLK Frequency

To change the ICLK frequency from less than 70 MHz to 70 MHz or higher such that the ratio of the frequency after the change to that before the change is greater than four, start by setting the frequency to one quarter of the intended frequency, wait for 3 μ s, and then change it to the intended frequency.

To change the ICLK frequency from 70 MHz or higher to less than 70 MHz such that the ratio of frequency after the change to that before the change becomes less than one quarter, start by setting the frequency to one quarter of the frequency before the change, wait for 3 μ s, and then change it to the intended frequency.

Examples of the respective procedures for changing the ICLK frequency are shown in Figure 9.16 and Figure 9.17.

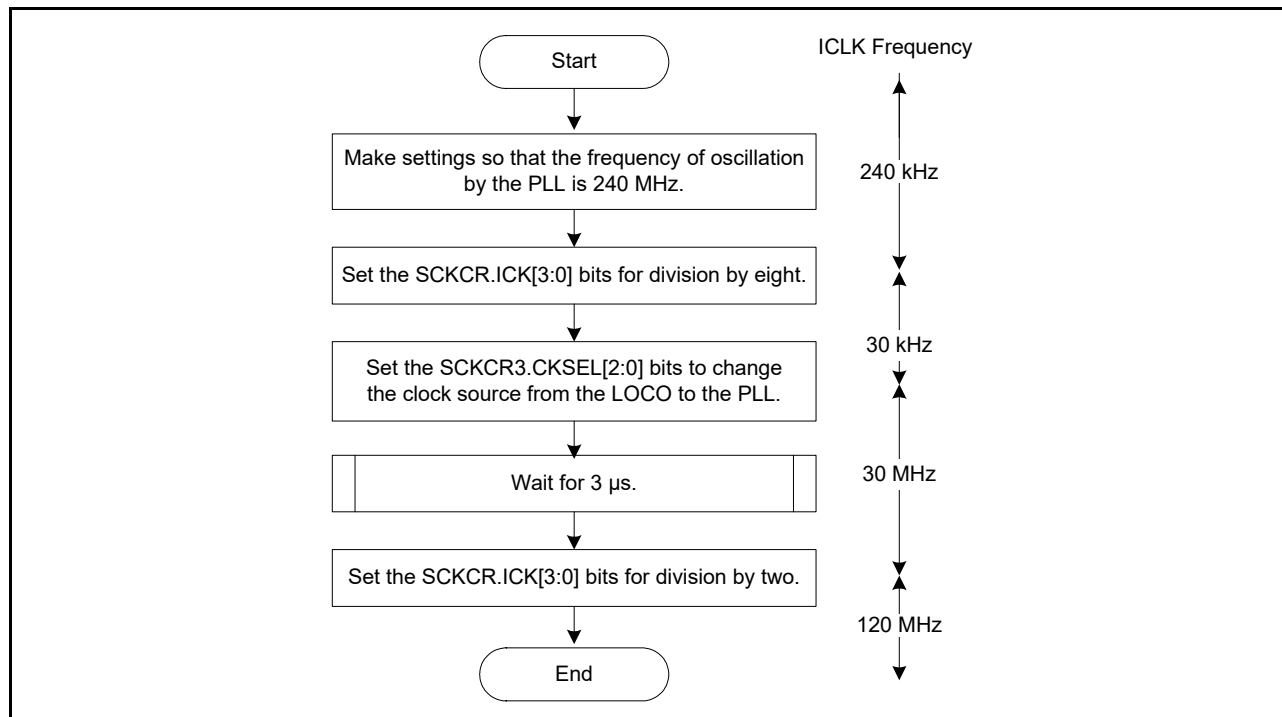


Figure 9.16 Example of Changing the ICLK from the LOCO Clock (240 kHz) to 120 MHz for the PLL Clock (240 MHz) through Division by Two

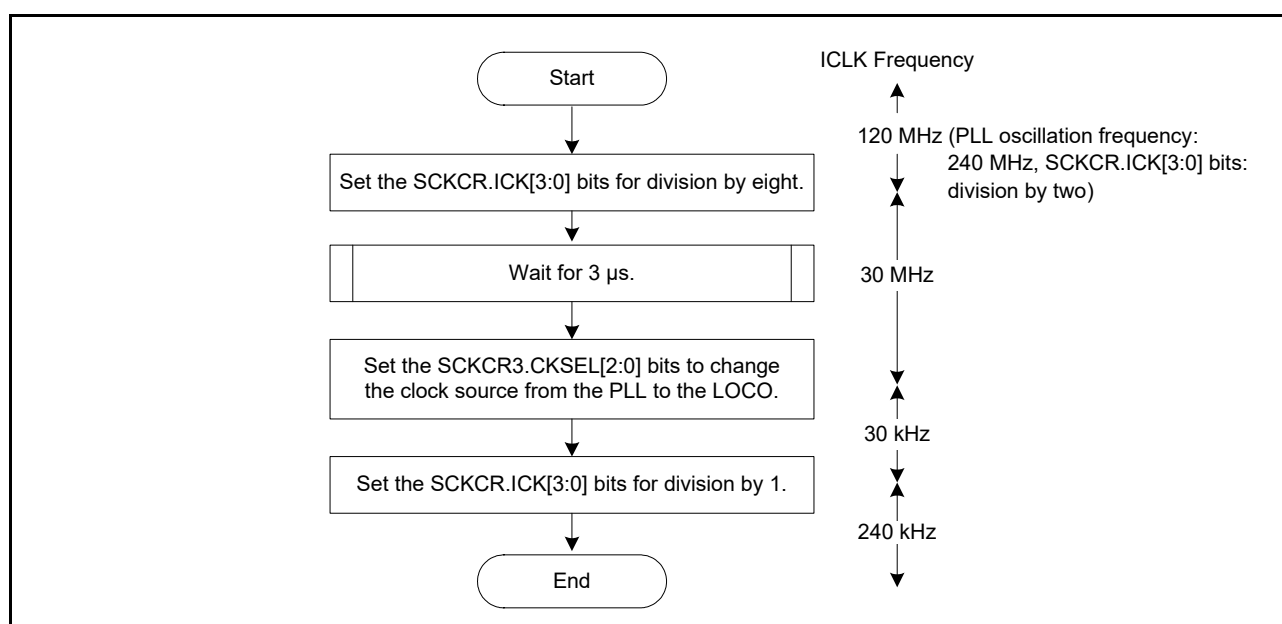


Figure 9.17 Example of Changing the ICLK from 120 MHz for the PLL Clock (240 MHz) through Division by Two to the LOCO Clock (240 kHz)